CS4222 Wireless Networking AY25/26 Sem 2. github.com/gerteck

1. Introduction to Wireless Networks

CS4222 Course Overview

- **Principles of wireless networking**: studied within the context of Internet of Things (IoT).
- Covers signal propagation, modulation, energy consumption estimation. Various layers of wireless networking stack (physical (PHY), MAC, networking, application). Case studies of popular wireless technologies (WiFi, Bluetooth), cellular networks, IoT.

Internet of Things (IoT)

- **IoT**: Network of embedded devices, sense, compute, communicate. **4 Core Tasks of IoT Device**: Compute and Storage, Sense and Actuation, Power Management, Wireless Networking.
- **Typical IoT Architecture**: Sensors and actuators at the edge (ADC: Analog to Digital Converter). Embedded processor for computation (MCU: Microcontroller Unit). Wireless interface for communication. Backend/cloud for storage and analytics.
- **ARM**: ARM (Advanced RISC Machines): low-power RISC (reduced instruction set computer) processor architecture (and ecosystem), widely used in embedded and IoT devices due to its energy efficiency, small footprint, and suitability for battery-powered wireless systems.
- **ARM Holdings Ltd.** provides the Instruction Set Architecture (ISA), CPU core designs (small part of the chip), tools etc. and licenses to semiconductor companies (e.g., Qualcomm, Apple, Samsung) to design and manufacture their own ARM-based chips.
- **Prevalence**: Billions of embedded devices, 4.4B Cortex-M processors in Q4 2020.
- **Energy Trend**: Computation cost low, Wireless comms energy expensive.

History of Wireless Communication

- **Early Wireless Milestones**: 1880: Photophone (sound over light), 1893: Wireless radio communication, 1899: km-scale radio transmission, 1901: Transatlantic wireless.
- **Broadcasting Evolution**: 1906: First AM radio audio broadcast. 1935: Frequency Modulation (FM) 1920s–30s: Radio stations worldwide. 1980s: Digital Audio Broadcasting (DAB)
- **Television Broadcasting**: 1923: First wireless TV transmit. 1941: NTSC standard (US). 1962: Telstar satellite. 1967: PAL standard (Analog TV). 2000s: Digital TV transition
- **WiFi Timeline**: 1985: ISM band opened for unlicensed use. 1997: IEEE 802.11 standard. 1999: 802.11a/b and WiFi Alliance. 1999: First consumer WiFi (Apple iBook)
- Every decade, lower prices, new usage, establish new industry.

Basics of Wireless Communication

- **Radio**: Latin *radius* (ray). EM waves (3 Hz to 300 GHz, collectively called **radio spectrum**).
- Radio technology implies transmitting and receiving EM waves using of frequency in radio spectrum, also known as radio waves.
- **Radio Transceivers**: for transmission/reception of EM waves. WiFi, Bluetooth, IoT devices
- **Electromagnetic (EM) Waves**: EM waves consist of oscillating electric and magnetic fields. Frequency: cycles per second (Hz). Period inversely proportional to frequency.
- EM waves travelling in free space, speed of light ($c \approx 3x10^8$ m/s). $\lambda = \frac{c}{f}$. ($v = fx$).
- **E.g. Wavelengths**: 98.7 MHz (FM) $\Rightarrow \lambda = 3$ m; 2.4 GHz (WiFi) $\Rightarrow \lambda = 0.125m$.

Radio Spectrum

- **Radio Waves**: Modern comms use radio waves, longest λ in EM spectrum (3 Hz to 300 GHz).
- **Radio Wave Characteristics**:
  - **Strength/Power**: energy contained in the radio wave.
  - **Frequency**: rate of periodic variation between electric and magnetic fields.
  - **Bandwidth**: range of frequencies occupied by the signal.
  - **Polarization**: orientation of the electric field - circular, elliptical, linear.
  - **Modulation**: method of encoding information onto the carrier wave.
- **Radio Wave Propagation**: Strongly influenced by environment. Obstacles cause reflection, diffraction, scattering. Non-line-of-sight common in real deployments
- **Practical Impacts**: Reduced communication range, Increased interference, Lower reliability

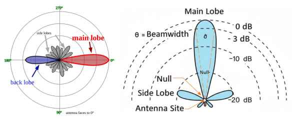
Band	Frequency Range	Wavelength Range	Applications
VLF	3 kHz – 30 kHz	100 km – 10 km	Maritime radio, nav
LF	30 kHz – 300 kHz	10 km – 1 km	Maritime, navigation
MF	300 kHz – 3 MHz	1 km – 100 m	AM radio, aviation radio, nav
HF	3 MHz – 30 MHz	100 m – 10 m	Shortwave radio
VHF	30 MHz – 300 MHz	10 m – 1 m	VHF TV, FM radio
UHF	300 MHz – 3 GHz	1 m – 10 cm	UHF TV, GPS, Wi-Fi, 4G
SHF	3 GHz – 30 GHz	10 cm – 1 cm	Satellites, Wi-Fi
EHF	30 GHz – 300 GHz	1 cm – 1 mm	Radio astronomy, satellite

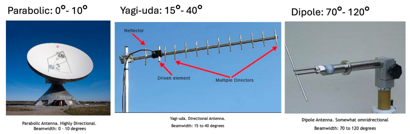
Radio Frequency Spectrum and Applications

2. Link Budget and Antenna Design

2.1 Antenna Basics, Directionality, Types

- **Radio Transceiver**: Electrical device that enables transmission and reception of EM waves.
- **Antenna**: Electrical conductor, transmit, receive EM waves. Converts electrical signals to radio waves and vice versa, reciprocal (equal performance for transmit TX and reception RX).
- **Types of Directionality**: directional, omnidirectional (radiates equally in all directions).
- **Common Antenna types**: Monopole, Array, Parabolic. Patch (flat element on substrate), loop, aperture etc. Dipole: Two conductors in opposite directions.





2.2 Antenna Size and Frequency

Antenna size is **inversely proportional** to frequency of radio wave ($\lambda = c/f$)

- Higher frequency, smaller antenna size, Lower frequency, larger size
- Dipole: length of conductor is $\frac{1}{2}$ of wavelength of radio wave
- Half-Dipole: length of conductor is $\frac{1}{4}$ of wavelength of radio wave
- Antennas work efficiently by resonating w. radio waves, when size is fraction of wavelength.

2.3 Antenna Gain: Performance of Antenna

- **Gain of antenna:** Dimensionless ratio (unitless in linear form), commonly expressed in decibels (dB) or dBi. Measures ability of antenna to focus energy in a particular direction compared to a reference.
  - **dBi**: Gain referenced to a theoretical isotropic antenna (uniform radiation in all directions)
  - **dBd**: Gain referenced to a half-wave dipole antenna ($dBd = dBi - 2.15$)
  - **Note**: Mathematically, dB and dBi treated same in a link budget, but conceptually they defer: dB represents actual loss or gains, dBi models antenna performance due to directionality.
- **Units:**
  - Linear gain (G) is unitless, converting to decibel scale to compresses ratios.
  - Logarithmic gain (G_{dB} or G_{dBi}) is in decibels (dB)
  - Convert: $G_{\mathrm{dB}} = 10\log_{10}(G_{\mathrm{linear}})$ and $G_{\mathrm{linear}} = 10^{G_{\mathrm{dB}}/10}$
- **When to convert:**
  - Use **linear gain** in physical formulas (e.g., Friis, $G = \frac{4\pi A_e}{\lambda^2}$), can convert to dBi after.
  - Use **dBi/dB** when doing link budget calculations (additive terms with power in dBm, FSPL in dB, etc.)
- $\uparrow$ gain $\Rightarrow$ more directional transmission/reception $\Rightarrow$ narrower beamwidth

$$\boxed{G = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi f^2 A_e}{c^2}}$$

- G = antenna gain (unitless)
- A_e = effective aperture area (m^2)
- f = carrier frequency (Hz)
- c = speed of light 3×10^8 m/s
- λ = carrier wavelength (m)

- **Directional Dependence:** Both G and A_e are functions of direction. The value used in link budgets is typically the **peak value** (main lobe).
- **Effective Area (A_e):** Represents the 'power-capturing' cross-section.
- Power output evaluated in a direction, relative to isotropic antenna output.
- **Array Gain**: $G_{\text{array}} = G_{\text{single element}}(\text{dBi}) + 10\log_{10}(N)$, where N is the number of antenna elements in array; $10\log_{10}(N)$ is the array factor. Multiply unitless gain also works.

2.4 Radiation Pattern and Beamwidth

- **Radiation Pattern**: graphical representation of relative strength of radiated or received power in various directions. 3D or 2D plots.
  - **Main Lobe**: direction of maximum radiation/reception.
  - **Side Lobes**: smaller lobes representing radiation/reception in other directions.
  - **Back Lobe**: lobe opposite to main lobe.
  - **Nulls**: directions with minimal radiation/reception.
- **Beamwidth**: angular separation between the $-3, \mathrm{dB}$ points (half-power points) of the main lobe. It defines the directivity and coverage area of the antenna.
  - $-3, \mathrm{dB}$ **points**: points on the radiation pattern where power drops to half of maximum value. ($10\log_{10}(0.5) \approx -3, \mathrm{dB}$).
  - **E.g. Antennas**: parabolic (highly directional), Yagi-Uda (directional), dipole antennas.

Antenna Beamwidth Tradeoffs:

- **Narrow beamwidth**: Offers higher directivity and longer range due to increased gain, making it suitable for point-to-point links.
- **Broader beamwidth**: Covers a larger area, broader coverage, but has lower directivity and range due to lesser gain, making it suitable for broadcast applications.

Line of Sight (LoS) vs Non-Line of Sight (NLoS) Propagation

Waves suffer attenuation due to distance, obstacles, interference. Underwater favors sound over radio waves. **Multipath**: signal takes two or more paths from transmitter to receiver.

- **Line-of-Sight (LoS) Propagation**: clear, direct, unobstructed path. Common in open areas.
- **Non-Line-of-Sight (NLoS) Propagation**: obstacles, reflection, diffraction, scattering. Common in urban, indoor environments. Reduces reliability and range.

2.5 Attenuation and Gain: Decibel (dB) and dBm Units

- **Attenuation:** Loss of energy when signal traverses through a medium
- **Gain:** $\uparrow$ in energy $\rightarrow$ devices employed to improve signal strength (E.g. amplifiers, antennas)
- **Decibels (dB)**: logarithmic unit to express ratios of power levels. We use decibels to express loss/attenuation of signal and gain. (express power ratios)
- $\boxed{dB = 10\log_{10}\frac{P_2}{P_1}}$ (where P_1 is input power, P_2 is output power)
  - e.g. signal travels through medium, power $\downarrow$ to $\frac{1}{2}$ original value, (i.e. $P2 = \frac{1}{2}P1$).
    $\implies 10\log_{10}\frac{P_2}{P_1} = 10\log_{10}0.5 = 10(-0.3) = -3dB$
- -3dB change is lose half signal power; $+10$dB is $10\times$ increase, 20dB is $100\times$.
- Use dB to measure changes in signal strength, as decibel values can be added and subtracted instead of multiplied and divided. E.g. signal experiences a $-3, \mathrm{dB}$ loss followed by a $+10, \mathrm{dB}$ gain, net change is $-3 + 10 = +7, \mathrm{dB}$, power increase of about $5\times$.

Unit of power of a Signal: dBm (Decibel milliwatt)

- Decibel milliwatt (dBm) is a measure of signal power.
- $\boxed{P(\mathrm{dBm}) = 10\log_{10}(P_{\mathrm{mW}})}$ (where P_{mW} is power in milliwatts)
- $0\ dBm$ represents a transmit power of $1mW$ $\rightarrow$ common reference point in comm. systems
- $30\ dBm$ corresponds to transmit power of $1W$, often max permissible output power for devices in unlicensed frequencies bands, e.g. WiFi, Bluetooth.
- E.g. Calculate power of signal with $dBm = -30$.
  $-30 = 10\log_{10}(P_{\mathrm{mW}}) \implies P_{\mathrm{mW}} = 10^{-3}\mathrm{mW} = 0.001\mathrm{mW}$.
- Key values: $0\mathrm{dBm} = 1\mathrm{mW}$. $20\mathrm{dBm} = 100\mathrm{mW}$. $30\mathrm{dBm} = 1\mathrm{W}$.

2.6 Estimating received Signal Strength: Friis propagation model

$$\boxed{P_r = G_r G_t \left(\frac{c}{4\pi f_c d}\right)^{\alpha} P_t}$$

(in dBm)

- f_c center frequency in Hertz
- c = speed of light
- d = distance btw transmitter/receiver
- a = path loss component
- G = antenna gain
- P = power received/transmitted

Environment	Path Loss Exponential (α)
Free Space	2
Urban area cellular radio	2.7 to 3.5
Shadowed urban cellular radio	3 to 5
In building LOS	1.6 to 1.8
Obstructed in building	4 to 6
Obstructed in factory	2 to 3

Wireless Signal Propagation Calculations

- **Friis Propagation Model**: Estimates received signal strength and range. Assumes LoS, free-space conditions, no obstacles or interference.
  - **Friis Transmission Equation**: $P_r = G_r G_t \left(\frac{c}{4\pi f_c d}\right)^{\alpha} P_t$. (unitless gains)
  - the term $\left(\frac{\lambda}{4\pi d}\right)^{\alpha}$ captures power reaching receiver, aka path loss factor. FSPL isolates and inverts this term to quantify signal attenuation (loss) over distance in free space.

2.7 Free Space Path Loss $FSPL(\textbf{linear}) = (\frac{4\pi df}{c})^2$

- Loss of power by signal as it transmits from the transmitter to receiver
- d is the distance between the transmitter and receiver antennas (in meters),
- f is the frequency of signal (in Hertz), c is speed of light in a vacuum
- $\boxed{FSPL = 20\log_{10}(d_{\mathrm{m}}) + 20\log_{10}(f_{\mathrm{Hz}}) + 20\log_{10}(\frac{4\pi}{c})}$ — in dB always
- Represents absolute minimum loss any wireless link will experience, real channels always worse. FSPL helps quantify the cost of moving to different frequencies and distances.
- **FSPL Formula Forms**:
  - FSPL$(dB) = 32.44 + 20\log_{10}(d_{\mathrm{km}}) + 20\log_{10}(f_{\mathrm{MHz}})$
  - FSPL$(dB) = 92.45 + 20\log_{10}(d_{\mathrm{km}}) + 20\log_{10}(f_{\mathrm{GHz}})$ (equivalent forms)
- Doubling distance or frequency each on own increases loss by $6, \mathrm{dB}$ (inverse square law).

2.8 Total link gain in wireless communication

- Net amplification/attenuation of EM signal along the communication path. Antennas, FSPL, any other system component is included.
  - $\boxed{(\text{dB}) = G_{Tx} + G_{Rx} - L_{Path} - L_{Misc}}$
  - G_{Tx} = Transmitter antenna gain (dB)
  - G_{Rx} = Receiver antenna gain (dB)
  - L_{Path} = Path loss (dB), including free-space loss
  - L_{Misc} = misc losses (dB), (atmospheric/cable losses/interference)

2.9 Link Budget, Link Margin, Total Link Gain

- **Link Budget**: Calculate received signal strength given all gains/losses in comm. system
- Total Link Budget (P_{rx}) = P_{tx}+ Total Link Gain
- **Total Link Budget** $\boxed{P_{Rx} = P_{Tx} + G_{Tx} + G_{Rx} - L_{Total}}$
  - P_{Tx} = Transmit power in dBm.
  - G_{Tx} = Transmit antenna gain in dBi.
  - G_{Rx} = Receive antenna gain in dBi.
  - L_{Total} = Total losses (e.g., free space path loss, cable losses) in dB.
  - P_{Rx} = Received power in dBm.
- **Link Margin**: Received power $-$ receiver sensitivity (**Signal Sensitivity**). (Represents buffer excess above bare minimum needed)
- **Google Project Loon Case Study**: Balloons at 20km altitude vs. satellites at 400km. Lower latency/power. -100 dBm sensitivity device comms balloon 20km away, only 5.3mW power.

2.10 dB, dBm and dBi operations

- **db** (Decibel): log unit for ratios (gain/loss), dimensionless multiplier, no abolute meaning
- **dBm**: absolute power reference to $1mW$, $0dBm = 1mW$ (physical quantity)
- **dBi**: Antenna gain relative to isotropic radiator. (directionality)
- **Adding dB values:**
  - If 2 components contribute gain/loss, we add their dB values
  - (logarithms turn multiplication into addition) $\rightarrow$ e.g. 2 amplifiers, +10dB, +5dB gain
- **Adding dBi to dBm to get EIRP:**
  - **Effective Isotropic Radiated Power (EIRP)**: total output power w. main lobe gain.
  - $\boxed{EIRP(\text{dBm}) = P_{TX}(\text{dBm}) + G_{antenna}(\text{dBi}) - L_{\text{cable}}}$
  - **Concept**: It treats the *max main lobe gain* as if it were radiated *omnidirectionally*.
  - **Note**: 'Theoretical' power. The actual total energy leaving the antenna is still P_{TX}.
- **Subtracting Losses (dB) from Power (dBm)**
  - Losses (cable/FPSL) can be subtracted from transmitted power
- **Subtracting dBm values to get dB ratio difference:** Used to compare 2 power EIRP. (E.g. $30dBm - 20dBm \rightarrow 10dB$ difference in power, note output unit is dB, not dBm)
- **Cannot add two dBm values directly**: Log values cannot be added directly
  - Must convert to linear scale e.g. $10dBm = 10mW$. $20dBm = 100mW$, Total power = $110mW = 10\log_{10}(110) = 20.4dBm$

Operations with dB, dBm, and dBi

Operation	Valid?	Result	Explanation
dBm + dBm	$\times$	Error	Log scale absolute powers cannot be added directly.
dBm - dBm	✓	dB	**Link Margin**: The ratio/gap between two power levels.
dBm + dBi	✓	dBm	**EIRP**: Absolute power increased by antenna directional gain.
dBm $\pm$ dB	✓	dBm	Power level adjusted by gains (amplifiers) or losses (cables/air).
dB + dB	✓	dB	Cumulative gain or loss (e.g., Cable loss + Connector loss).
dBm to linear mW	✓	mW	$P_{mW} = 10^{(P_{dBm}/10)}$
Linear mW to dBm	✓	dBm	$P_{dBm} = 10\log_{10}(P_{mW})$
$\therefore$ dBm + dBm (via linear)	✓	dBm	Convert to mW, sum, then log: $10\log_{10}(10^{P_1/10} + 10^{P_2/10})$

Frequency Band and Antenna Reference

Band / Technology	Freq.	Wavelength (λ)	$\lambda/2$ Dipole	Primary Use Case
LTE Band 28 (Low)	700 MHz	42.9 cm	21.4 cm	**Coverage**: Deep indoor penetration and rural long-range.
LTE Band 3 (Mid)	1800 MHz	16.7 cm	8.3 cm	**Capacity**: Standard urban 4G data layer.
WiFi 2.4 GHz	2400 MHz	12.5 cm	6.25 cm	**Compatibility**: Wide range but high interference (crowded).
5G Sub-6 (n78)	3500 MHz	8.6 cm	4.3 cm	**Performance**: The 'Sweet Spot' for high-speed 5G mobile.
WiFi 5 GHz	5000 MHz	6.0 cm	3.0 cm	**Throughput**: Fast home networking; poor wall penetration.
5G mmWave (n257)	28 GHz	10.7 mm	5.4 mm	**Ultra-Speed**: Gigabit speeds in stadiums/hotspots; very short range.

- **Resonance:** Antennas most efficient at $\lambda/2$, current, voltage waves reinforce each other.
- **Trade-off:** Lower frequencies (700 MHz) provide better coverage but require physically larger antennas that are harder to fit in phones.

3. Modulation Techniques for Wireless Communication

How are radio waves modified to carry information

Radio waves are pure sine waves that carry no information. To modify to send info:

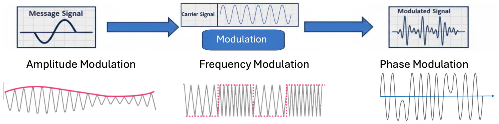
- Start with low frequency baseband signal (message signal)
- Use it to modify a high frequency carrier signal (radio wave)
- Transmit modified carrier signal over air, receiver demodulates to recover baseband signal

This process is called **modulation**. Modulation Techniques include Amplitude Modulation (AM), Frequency Modulation (FM), Phase Modulation (PM), or vary them together and advanced digital schemes like QAM, PSK.

3.1 Modulation

Encoding 'baseband' information into analog (continuous, smooth waveform)'carrier' signal. Baseband signal often $\downarrow$ frequency than carrier. Frequency of carrier dictates spectrum usage.

- **Why use a high-frequency carrier instead of sending baseband directly?**
  - Baseband signals or info change at a much lower rate. (E.g. Speech up to 20KHz, temp changes $\approx$ few Hz) - these signals are slow compared to radio waves (MHz to GHz).
  - If transmit baseband directly, need large antennas (size $\approx \frac{1}{4}\lambda$), and frequency overlaps.
- **Benefits of modulating onto high frequency carrier**
  - More efficient (smaller) antenna size, better radio spectrum usage. (e.g. 2.401 GHz, 2.412 GHz - frequency division multiplexing).



- **Modulation Index** μ, β: Extent carrier signal is modulated w. message signal. Determines strength of modulated signal, impacts bandwidth, power distribution, overall signal quality.

3.2 Amplitude modulation (AM)

- Change amplitude of carrier signal according to the message signal: $s(t)$: modulated signal

$$\boxed{s(t) = [A_c + A_m \cdot m(t)]\cos(2\pi f_c t)}$$

  - A_c Carrier amplitude
  - A_m Message (baseband) amplitude
  - $m(t)$ Message signal
  - f_c Carrier Frequency
- **Signal (Single Tone)**: $s(t) = A_c[1 + \mu\cos(2\pi f_m t)]\cos(2\pi f_c t)$
- **Multi-tone Signal**: $s(t) = A_c\left[1 + \sum_{i=1}^{n}\mu_i\cos(2\pi f_{m_i} t)\right]\cos(2\pi f_c t)$
- **Product-to-sum Id**: $\cos A\cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$, to derive sidebands
- **Sidebands**: $f_c \pm f_m$. Information-carrying frequencies produced above (USB)/below (LSB) carrier. 2 sidebands per message tone. Standard AM power-inefficient, carrier $\approx 67\%$ power.
- **AM Modulation Index** μ
  - For amplitude modulation $(modulation\ index): \mu = A_m/A_c$
    - A_m = Peak amplitude of modulating/baseband signal
    - A_c = Peak amplitude of carrier signal
    - For AM: $\mu < 1$: under-modulated (weak signal, no distortion); $\mu = 1$: fully modulated; $\mu > 1$: over-modulated (distortion, clipping)
  - **Multi-tone Modulation Index** (μ_{total}): message with multiple frequencies with individual indices $\mu_1, \mu_2, \ldots, \mu_n$: $\mu_{total} = \sqrt{\mu_1^2 + \mu_2^2 + \cdots + \mu_n^2}$
- **AM Power Analysis**
  - Carrier Power: $P_c = \frac{A_c^2}{2R}$ (R: antenna resistance Ω, i.e. sinusoidal signal power)
  - Sideband Power: $P_{sidebands} = P_{sb} = P_c \cdot \frac{\mu_{total}^2}{2} = \frac{A_c^2}{2R} \cdot \frac{\mu_{total}^2}{2}$
  - *Direct Calc*: $P_{sb} = \sum_{i=1}^{n}\left(2 \cdot \frac{A_{sb,i}^2}{2R}\right)$. The $\times 2$ accounts for the **pair** (USB + LSB).
  - Total Transmitted Power: $\boxed{P_t = P_c\left(1 + \frac{\mu_{total}^2}{2}\right)}$
  - Efficiency (η): $\eta = \frac{P_{sidebands}}{P_{total}} = \frac{\mu_{total}^2}{2+\mu_{total}^2}$
- **AM Spectrum and Bandwidth**
  - **Frequencies present**: Carrier (f_c), Lower Sideband ($f_c - f_m$), Upper Sideband ($f_c + f_m$). (LSB/USB: Find using $f_m = \text{freq}_{component} - f_c$)
  - Bandwidth: $\boxed{BW_{AM} = 2 \cdot f_{m,max}}$ *(twice highest frequency component of message)*

3.3 Frequency modulation (FM)

- Change frequency of carrier signal according to the message signal
- Advantages over AM: resilient to interference, spectrum efficiency, used for radio
- **Bandwidth**: how much 'room' signal occupies in frequency spectrum. FM bandwidth depends on how *fast* and *how much* you wiggle carrier freq around f_c to carry information.

$$\boxed{s(t) = A_c cos(2\pi f_c t + 2\pi k_f \int m(\tau)d\tau)}$$

  - E.g. $m(t) = A_m\sin(2\pi f_m t)$: $s(t) = A_c\cos(2\pi f_c t - \beta\cos(2\pi f_m t))$
  - A_c Carrier amplitude, f_c Carrier Frequency
  - k_f Frequency sensitivity of modulator (Δ carrier freq. to Δ amp of modulating signal)
  - $m(t)$ Modulating signal $\implies f_m$ (signal max frequency), A_m (max amplitude)
- **FM Modulation Index:** β (in Radians, Peak Phase Deviation)
  - $\beta(modulation\ index) = \Delta f/f_m$, where $\Delta f = k_f \cdot A_m$ (Peak frequency deviation)
  - $\beta < 1$: Narrow band signal, $\beta > 1$ Wideband (WBFM, $\uparrow$ noise immunity, signal quality).
  - Given by Carson's rule, bandwidth $B \approx 2(\Delta f + f_m)$
- **Significant Sidebands**: FM produces infinite sidebands at $f_c \pm nf_m$ (for $n = 1, 2, \ldots$), but only those with $\geq 1\%$ power are significant. Number of significant sideband pairs is approx $N \approx \beta + 1$. Total Spectral components = $2N + 1$ (sideband pairs + carrier).

3.4 Phase modulation

- **Phase**: position of point on waveform cycle, offset from starting position of sine wave
- Complete cycle of sine wave is 360 degrees, i.e. 2 π radian

$$\boxed{s(t) = A\cos(2\pi ft + \phi)}$$

  - A = Amplitude of signal
  - f = Frequency of signal
  - ϕ Phase (phase contstant / offset)

- ϕ determines where in cyclewave (begins at $t = 0$), $\Delta\phi$ is phase shift left/right in time
- **Impact of phase shifts on waves (summation)** (interference)
  - *Constructive:* 2 waves with same phase add tgt to form larger amplitude
  - *Destructive:* 2 waves out-of-phase (e.g. 180° apart), cancel each other

Phase Modulation (PM)

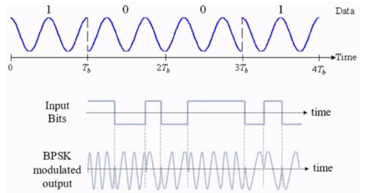
- Change phase of carrier signal according to the message signal
- Advantages over FM: Better SNR and spectral efficiency

$$\boxed{s(t) = A\cos(2\pi f_c t + 2\pi k_p m(t))}$$

  - A = Amplitude of signal
  - f_c = Carrier frequency
  - k_p = Phase sensitivity (rad / unit amplitude)
  - $m(t)$ = modulating signal

Binary Phase Shift Keying (BPSK)

- simple form of phase modulation where phase is shifted between 2 values
- *Bit 0* $\rightarrow$ 0° ($A\sin(2\pi f_c t)$); *Bit 1* $\rightarrow$ 180° ($A\sin(2\pi f_c t + \pi)$)



Quadrature Phase Shift Keying (QPSK)

- extends BPSK with 4 distinct (closer) phase states, allow encoding of 2 bit/symbol.
- *Bit 00* $\rightarrow$ 45° ($A\sin(2\pi f_c t + \pi/8)$, *Bit 01* $\rightarrow$ 90° ($A\sin(2\pi f_c t + \pi/4)$)
- *Bit 10* $\rightarrow$ 135°, *Bit 11* $\rightarrow$ 225°. *QPSK vs. BPSK*: $2\times$ bit rate for same bandwidth.

3.5 Traditional Digital Modulation (ASK, FSK, PSK)

Core Concept: Map binary bits (0s, 1s) onto carrier wave, 'shift' physical discrete states.

- **ASK (Amplitude Shift Keying)**: Varies **Signal Strength**.
  - 1 = Wave Present (High Amplitude); 0 = No Wave (Low/Zero Amplitude).
  - *Usage*: E.g. OOK - On-Off Keying, good for short-range LoS, low interference, simplicity
- **FSK (Frequency Shift Keying)**: Varies **Signal Speed**.
  - 1 = High Frequency (tight waves); 0 = Low Frequency (spread waves).
  - *Usage*: Resistant to noise; medium complexity (low-power system), sufficient range.
- **PSK (Phase Shift Keying)**: Varies the wave's **Timing/Starting Point**.
  - Bits are represented by 'jumps' or phase reversals (e.g., a $180°$ flip to signify a bit change).
  - E.g. Chirp Spread Spectrum LoRa, robust to interference, long-range, highest complexity
- **Significance**: Translation layer: Computers binary, antennas continuous analog waves.

4. Channel Capacity and Link Quality

Wireless Link Quality and Reliability Metrics:

- **Packet Delivery Ratio (PDR)**: ratio of successfully received packets to transmitted packets.

$$\boxed{PDR = \frac{Received\ Packets}{Transmitted\ Packets} \times 100}$$

  - ↑ = better link quality, ↓ = congestion/interference or poor connectivity
- **Bit Error Rate (BER):** Ratio of bits received in error to the total transmitted bits. Indicates transmission quality at the physical layer.

$$\boxed{BER = \frac{Errored\ Bits}{Total\ Bits\ Transmitted} \times 100}$$

  - ↓ = fewer errors + better link perf. ↑ = data corruption / retransmissions
  - Wired Ethernet: 10^{-12}, Wireless Wifi: 10^{-6} to 10^{-3}, Poor cxn = 10^{-2}
- **BER-PDR Relationship**: PDR depends on BER (p) and packet length (n).
  - *Exact*: $PDR = (1-p)^n$ (All n bits must be correct for packet success).
  - *Approx*: $PDR \approx e^{-np}$ (Valid for small p, large n; np is the avg. errors/packet).
- **Signal-to-Noise Ratio (SNR):** Ratio of signal power to background noise, often measured in decibels (dB). $SNR_{dB} = 10\log_{10}(SNR_{linear})$.

$$\boxed{SNR = 10\log_{10}\left(\frac{P_{signal}}{P_{noise}}\right) \text{— if } P \text{ in } W}$$

  - Else $P_{signal} - P_{noise}$ in dBm
  - ↑ = stronger signal, fewer errors. ↓ = more packet loss.
  - SNR $> 25dB$ = excellent (4G LTE etc). SNR $< 10dB$ = unreliable link.
  - **Noise Power Spectral Density (N_0):** Noise power per bandwidth ($P_n = N_{0,\text{W/Hz}} \cdot B$);
  - **Noise Power (P_n):** $P_n = 10\log_{10}(N_{0,\text{dBm/Hz}} \cdot B)$. ($\log_{\text{dBm/Hz}}$; $\text{linear}_{W/Hz}$ units)
  - **Thermal Noise floor**: $N_0 = kT$, at room temp, $N_0 \approx -174$ dBm/Hz.
- **Expected Transmission Count (ETX):** Measures expected number of trasmissions (including re-trasmission) required for packet to successfully received. Used in routing algorithms.

$$\boxed{ETX = \frac{1}{P_{forward} \times P_{reverse}} = \frac{1}{PRR(\text{Packet Reception Rate})}}$$

  - Lower ETX means fewer transmission needed per packet and better efficiency, higher leads to increased latency and power consumption. P_{forward}: forward packet delivery probability.

Understanding Channel Capacity & Link Quality

- **Channel Capacity:** Max rate information can be transmitted over communication channel without error, measured in **bits/second**.
- How does channel capacity relate to link quality?
  - **Higher SNR**: higher capacity, can distinguish subtle phase/amplitude Δ, more bits/symbol.
  - **Bandwidth** as frequency range signal transmitted; wider bandwidth allows more data to fit, higher transmit rates.
  - **Link Quality Factors**: Distance (path loss), Fading, Obstacles (shadowing) interference, reduce channel capacity.
- **Shannon's Channel Capacity Formula** (Theoretical maximum)

$$\boxed{C = B \times \log_2\left(1 + \frac{S}{N}\right)}$$

  - C Channel capacity in bits per second (bps)
  - B = Bandwidth of the channel in Hertz (Hz)
  - S/N Signal-to-noise ratio (SNR) expressed as linear ratio (convert: $10^{\text{SNR(dB)}/10}$)

Channels, Subchannels and Overall Wireless Link Quality

- Total bandwidth divided into **subchannels**, each w. own bandwidth (B_i), SNR (S/N_i).
  - Capacity per subchannel: $C_i = B_i \log_2(1 + S/N_i)$
  - **Overall Channel capacity**: sum of capacities of all subchannels.

$$\boxed{C_{\text{total}} = \sum_i C_i = \sum_i B_i \log_2\left(1 + \frac{S}{N_i}\right)}$$

  - Intuition: Some parts of the channel may experience more noise or interference than others. Treating them separately allows a more accurate capacity estimate.
  - Capacity is theoretical upper bound. Real-world factors (protocol overhead, interference, fading) reduce actual throughput below limit. Link quality assessments (PDR, ETX, BER) provide practical insights into how close to theoretical capacity in real deployments.

Link Quality and Channel Capacity Relationship

- **Link quality** is determined by reliability (PDR, ETX), physical-layer conditions (BER, SNR).
- **High PDR/SNR, low ETX/BER** : reliable, efficient packet delivery, fewer retransmissions.
- **Poor link conditions** (fading, interference, noise) reduce effective capacity even if theoretical capacity is high. In practice, **wireless systems aim to maximize SNR and minimize interference** to approach theoretical capacity limits.

5. Sensors and Actuators

5.1 Sensor Modeling

- **Definition**: Components measuring physical phenomena.
- **Affine (linear r/s) Model**: $f(x(t)) = ax(t) + b$
  - $x(t)$: True physical quantity; $f(x(t))$: Reported value.
  - **Sensitivity** (a): Change in reading per unit physical change.
  - **Bias** (b): Offset/reading when true input is zero.
- **Inversion (Calibration)**: To recover physical value from raw reading: $x(t) = \frac{f(x(t))-b}{a}$.
- **Saturation**: Real sensors have limited ranges $[L, H]$. Outside this, readings 'saturate' (clip to max/min), and the affine model fails. (E.g. below min, report garbage/min value; above max, report max or err no reading).
- **Calibration Errors**: Gain error (wrong a) i.e. systematic scaling error, Offset error (wrong b) i.e. consistent bias.

5.2 Electrical Transduction

- **Mechanism**: Converting phenomena into electrical signals (Voltage/Current/Resistance).
- **Ohm's Law**: $V = I \times R$ where $R = \frac{\rho L}{A}$.
- **Voltage Dividers**: Essential for converting variable Resistance into measurable Voltage (V_{out}) for an ADC. $V_{out} = V_{in} \cdot \frac{R_2}{R_1+R_2}$
- **Resistivity** (ρ): Changes based on physical conditions:
  - **Temperature**: Conductors (higher ρ at higher T); Thermistors (non-linear).
  - **Light**: Photoresistors/Photocells (non-linear ρ change).
  - **Strain**: Strain gauges (ρ increases when stretched).
- **Piezoelectric Effect**: Some sensors (accelerometers, ultrasound) generate voltage directly via mechanical compression/strain rather than changing resistance.

5.3 Analog-to-Digital Conversion (ADC)

- **Function**: Bridges analog (continuous) and digital (discrete) domains.
- **ADC Parameters**:
  - **Resolution**: Number of bits. More bits = finer granularity.
  - **Sampling Rate**: Frequency of samples. Must follow **Nyquist**: $f_s \geq 2 \times f_{max}$.
  - **Accuracy**: Closeness to true value (impacted by noise and linearity).
- **Dynamic Range** (D): Ratio of range to precision p.
  - $D = \frac{H-L}{p}$ or $D_{dB} = 20\log_{10}(\frac{H-L}{p})$, Precision $p = \frac{H-L}{2^{bits}-1}$.
- **Quantization**: Mapping continuous $x(t)$ to one of 2^n discrete digital values.
- **Hardware Trade-off**: For a fixed cost, faster ADCs (high f_s) typically produce fewer bits, leading to higher quantization error or smaller range.

5.4 Sampling and Time

- **Sampling Signal Model**: Maps a physical quantity $x(t)$ to a discrete signal $s: \mathbb{Z} \to \mathbb{R}$.
  - **Function**: $s(n) = f(x(nT))$ where $x(t)$ is observed only at discrete times $t = nT$.
  - **Sampling Interval** (T): Fixed time between samples.
  - **Sampling Rate**: $f_s = 1/T$, measured in Samples/second or Hz.
- **Uniform Sampling**: Discrete signal $s(n) = f(x(nT))$ where T is the sampling interval.
- **Trade-offs**: Smaller T (higher rate) increases ADC cost/power and usually reduces bit-depth (resolution) for the same hardware cost.
- **Noise**: Undesired signal component $n(t) = x'(t) - x(t)$: quantization err/imperfections.

5.5 Specific Sensor Types

- **Active vs. Passive**: Passive sensing (e.g., PIR) is cheaper and lower energy. Active (e.g., Ultrasound) consumes more power.
- **Motion (Accelerometer)**: Measures 'proper acceleration' by mass displacement. Units in g.
  - **Model**: $f: (L, H) \to \{0, \ldots, 2^b - 1\}$ maps acceleration range to discrete bits.
  - **Tilt Sensing**: Uses gravity to find angle θ: $A[x] = 1g \cdot \sin(\theta)$.
  - **Localization**: Can estimate position $p(t)$ by double integrating acceleration $x(t)$, but error is squared over time. Usually used to 'fill gaps' between GPS samples.
- **Rotation (Gyroscope)**: Detects orientation changes (MEMS based); unaffected by gravity.
- **Magnetic (Magnetometer)**: Detects Earth's magnetic field; used for compasses/localization.
- **Active (Ultrasound)**: Emits sound and measures Time-of-Flight (ToF). Distance $d = \frac{v \times t}{2}$. Power-hungry but useful for proximity.
- **Passive Infrared (PIR)**: Detects movement via changes in Infrared (heat) levels. Low energy.

5.6 Actuators

- **Definition**: Modifies physical quantities (opposite of sensors).
- **LEDs**: Directional (Cathode/Short lead = Negative).
- **RGB LEDs**: Combines 3 colors to create any spectrum.
- **Audio**: Speakers/Horns (e.g., 85 dB horn in Nest Protect).

6. Power & Backscatter Wireless Communication

6.1 Power Consumption

Battery capacity

- **Watt-hour (Wh):** unit of energy equivalent to 1 Watt of power expended for 1 hour of time
- **Ampere-hour (Ah):** unit of electric charge equal to charge transferred by steady current of 1A flowing for 1 hour
- **(Ah, mAh) not directly comparable**: only comparable if voltages are identical. E.g. to compare powerbank (mAh), calculate (Wh) with internal voltage to compare.
  True Energy: $Wh = (Ah \times V)$, $Wh = \frac{mAh \cdot V}{1000}$.
- **Power vs. Energy Formulas**:
  - **Instantaneous Power**: $P(Watts) = V(Volts) \times I(Amperes)$.
  - **Total Energy**: $E(Wh) = P(W) \times t(hours)$.
- **Lifespan Estimation**: $Lifetime(h) = \frac{Capacity(mAh)}{AverageCurrent(mA)}$ or $\frac{Energy(Wh)}{Power(W)}$.
- **Trade-off**: WiFi (high power, e.g., ≈ 16h on coin cell) is for data; Bluetooth/Low-power sensors (low power, e.g., > 200h) are for longevity.

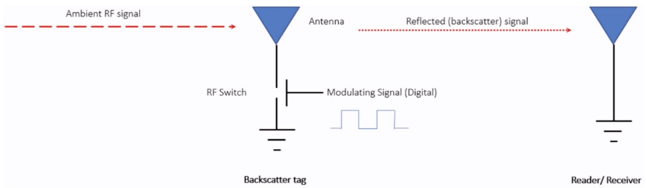
Device Power Consumption Optimization

- **Dominant Factor**: Radio comms (WiFi, Bluetooth) primary power drain for IoT devices, often exceeding the CPU or sensors.
- **Power States**: Radios use distinct states (*Transmit, Receive, Idle, Sleep*) to save energy. Optimization involves keeping the radio in *Sleep* mode as much as possible.
- **State Transition Overhead**:
  - Switching from Sleep to Active takes time and energy.
  - While low-level tasks (e.g., WiFi Beacons) transition quickly, heavy data communication is delayed by driver/software overhead (e.g. wake CPU, load network stack).
- **Optimization Goal** (extend battery life): Maximize 'Duty Cycling' (sleep to active time), min. transmit time/freq, data aggregation before transmit, reduce P_{TX} if range allows.

6.2 Low-power Wireless Transmission: Backscatter

Backscatter Communication

- Device does not generate its own power; instead, it varies between absorbing or reflecting signal to modulate data onto existing radio waves in the environment
  - Wireless transmission at microwatts of power draw (10000x energy savings)
  - Frequency bands: 400MHz, 900MHz 2.4GHz



6.3 Far-field Backscatter (Radiated Reflection)

- **RFID**: Radio Frequency Identification, widely deployed backscatter system. Cheap tags, complex readers (low/high/ultra frequency readers).
- **Active / Semi-Passive tags**: Active → own power source, semi-passive → battery for internal circuitry, rely on reader signal for communication. Greater range but bulky, high cost.
- **PushID [NSDI 2019]**: extend range without modify tag, leverage collaboration between multiple RFID readers. (beamforming)
- **Decouple signal generation and reception (LoRea)**: long range backscatter transmission. Dedicated transmitter generates strong signal, separate receiver captures weak backscatter signal. Range up to 3.4 km. Self-interference reduced.
- **Backscatter Usages**: ID, transmissions, backscatter as a sensor in gesture monitoring.
- Overall, backscatter is essentially 'free' communication (given a high capability device), but difficult to reflect already low energy (doubled path loss there and back), limited range.

6.4 Near-field Backscatter (Inductive Coupling)

Backscatter Communication using Inductive Coupling (Magnetic Induction)

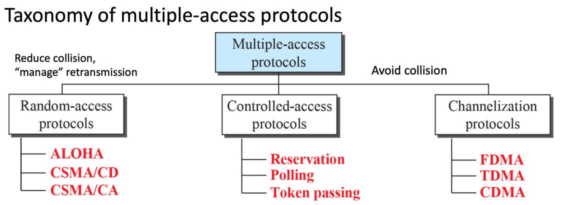
- Shared magnetic field created between 2 devices
- $\triangle$ current through one induces current charge on other, device can vary load to transmit
- Very low frequency bands (135 KHz, 13.56 MHz), transmit through materials like skin, but sensitive to metal

Near Field Communication (NFC)

- Inductive Coupling concept (13.56 MHz): attached to powered + capable device, $10-20$cm range, Data range $100-400$ kbps
- Act as tag/reader (smart phone can power a tag if needed, act as card to respond to reader)

7. Wireless Medium Access Control (MAC)

- Wired networks: reliable, physically isolated. Consistent throughput, low packet loss.
- Wireless: Lack of wires introduces variability, unreliability, vulnerability to interference
- Increasing **wireless network capacity** is challenging.



7.1 Channelization Protocols

Channelization: divide single communication medium into multiple distinct channels, transmit without collisions through preassign, but rigid allocation wastes capacity under bursty traffic.

TDMA — Time Division Multiple Access

- Split transmission in time, devices share same channel
- Splits time into fixed-length windows: devices assigned ≥ 1 windows, priority system
  - Devices listen periodically to resynchronize
  - Large guard windows e.g. 1.5 second slot for 1 second trans.
- **Pros/Cons**: Fixed time slots $\implies$ predictable latency, no collisions, efficient in centralized networks, but not dynamic, struggle in variable loads, dynamic device connections.

FDMA — Frequency Division Multiple Access

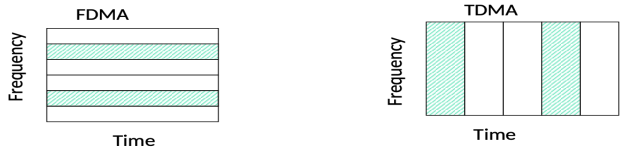
- Separate transmissions by placing then at diff freq. Transmit at same time.
- Technically, each device can use a separate, non-overlap, fixed frequency
  - E.g. walkie-talkies (mostly separate), $\uparrow$ throughput and reliability, spatial diversity
  - **Disadvantages:** If devices not always transmitting, part of spectrum is not used, inefficient.
- **WiFi** - Overlapping channels, pack more concurrent users (OFDMA).

Multi Channel MAC Protocols

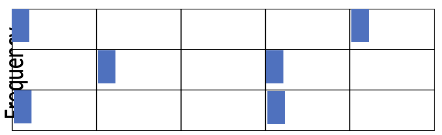
- Control channel: default, time syncrhonization for control/msg exchange
- Nodes switch to different channel for data communication
- All channels used for comm. , node switches channel when link quality bad

Orthogonal Frequency Division Multiplexing — OFDM

- FDM: entire frequency spectrum is dedicated to carry data from a single source
- OFDM: data divided into many (slower) bit streams and multiple subcarriers (each with smaller spectrum used)

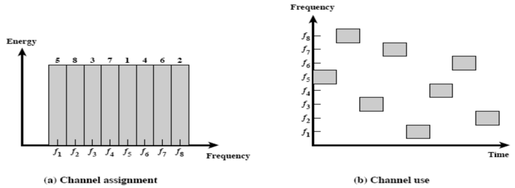


In practical systems, TDMA is often combined with FDMA



Frequency and Time leads to Spread Spectrum (SS)

- Spread-spectrum techniques are methods by which a signal is deliberately spread in the frequency domain, resulting in a signal with a wider bandwidth
- Such techniques can be used to establish secure communications, increase resistance to natural interference, jamming and to prevent detection
- **Advantages:**
  - Resistant to narrowband interference
  - Signals are difficult to interpret – appears as if increase in background noise
  - Can share frequency band with others with minimal interference



FHSS increase resistance to a) natural interference b) jamming c) detection

Spectral Efficiency

- Spectral efficiency (in bits/s/Hz) determines how efficiently devices utilize allocated bandwidth, depending on the modulation scheme
- Different modulation schemes provide different spectral efficiency:
  - Simple modulation (e.g., FSK) $\approx$ 1 bit/s/Hz
  - Advanced modulation (e.g., QPSK, QAM) higher bits/s/Hz (2–8 bps/Hz).

7.2 Random Access Protocols

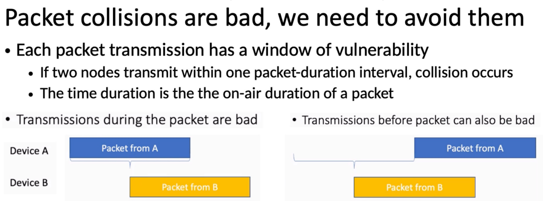
In Random Access Protocols (RAPs), users are allowed to transmit at any time without pre-scheduling, but risk collisions.

For dynamic wireless networks, fixed scheduling often inefficient or impractical:

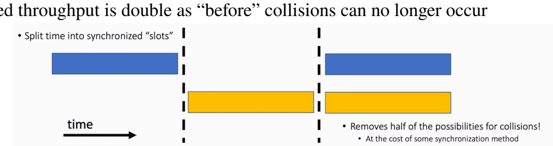
- Pre-allocate transmit slots inefficient when traffic is bursty/unpredictable, waste resources if nothing sent during allocated slots.
- RAPs are efficient, lightweight for managing sporadic, unscheduled, low-rate data transmits.

Aloha

- Devices transmit whenever data is available, without explicit coordination, no prior scheduling or channel sensing; purely random transmissions
- **Rules:**
  1. If you have data, transmit immediately
  2. After transmission, wait a certain time to receive acknowledgment (ACK)
  3. If ACK is received, transmission successful; if no ACK, wait a random time interval and retransmit (exponential backoff recommended)
- Collision Problem:
  - If ≥ 2 nodes transmit simultaneously, collision occurs, results in lost data
  - Nodes do not listen before transmitting (no channel sensing)



- **Slotted Aloha:**
  - Slotted ALOHA reduces collision windows by aligning transmissions to predefined slots
  - Requires sync: devices must know slot boundaries, add complexity to implementation
  - Slotted throughput is double as "before" collisions can no longer occur



- **ALOHA Throughput**: Shown that traffic maxes out at ALOHA throughput of 18.4% (slotted ALOHA doubles to 36.8%), assuming Poisson distribution of traffic.

Why Use Random Access if Efficiency is Limited

- *Simplicity and low overhead:*
  - RAPs do not need complex synchronization or scheduling mechanisms
  - Suitable for resource-constrained IoT or sensor networks
- *Efficiency in low-load scenarios:*
  - Collisions are rare when network load is low
  - Random access can achieve low latency for intermittent or bursty transmissions
- *Real-world usage examples:*
  - BLE beaconing: Short, random broadcasts to discover devices
  - LoRaWAN: Simplicity and battery efficiency preferred over maximizing throughput
  - Satellite and RFID networks: Sporadic data transmission scenarios

Carrier Sense Multiple Access w/ Collision Avoidance (CSMA/CA)

CSMA/CA medium access control protocol widely used in wireless networks (especially WiFi). Nodes sense medium to check if busy before attempting transmission, collisions actively avoided rather than detected after occurrence.

- Wireless networks have difficulty reliably detecting collisions directly due to signal fading, interference, and hidden nodes.
- Collision Avoidance proactively reduces the probability of collisions occurring.

CSMA/CA Algorithm Steps:

1. *Listen first (Carrier Sensing):* Nodes listen before transmission. If channel busy, node waits.
2. *Random Backoff (Collision Avoidance):* Once channel idle, nodes don't transmit immediately. Wait a random time interval (random backoff period) before transmitting.
3. *Transmit Data:* After random wait, if medium remains idle, node transmits its data.
4. *Acknowledgment (ACK):* Successful reception is confirmed through an ACK from the receiving node. Lack of ACK implies potential collision or error, triggering retransmission.

Choosing W_{max}: Maximum Backoff Window Size

- Intuitively, W_{max} should be small when lower chance of collision and vice versa
- W_{max} varies according to the Binary Exponential Backoff (BEB) algorithm.
- Set "slot time" to 2 times the max. propagation delay on the channel
- If this is the first transmission, sends immediately
- If there collision, after i-th collision, pick W_{max} randomly between 0 and $2^i - 1$ time slots.
- **Optimal point**: depends on no. of nodes condenting + traffic load. Too aggressive (small W_{max}): more collisions, too conservative (large W_{max}): underutilization of channel.

Binary Exponential BBackoff (BEB)

- If collisions occur, wait intervals adaptively increase to reduce future collision probability
- Each successive collision indicates higher channel contention
  1. Initial attempt: Small waiting period (randomized within a small range).
  2. After each collision: The range (window size) for the random backoff doubles (exponential increase), reducing collisions progressively:
     - 1st collision: Wait between $0 - (2^1 - 1)$ slots
     - 2nd collision: Wait between $0 - (2^2 - 1)$ slots
     - After i-th collision: Wait between $0 - (2^i - 1)$ slots

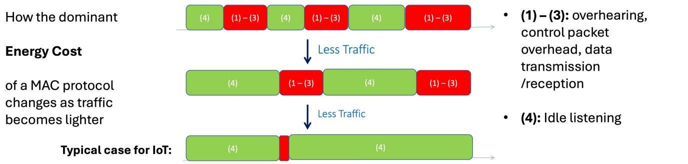
TCSMA with RTS/CTS (Threshold CSMA)

- **Hidden Terminal Problem**: Occurs when two transmitters cannot sense each others transmissions but interfere at a common receiver. (Not hearing another sender does not necessarily mean the receiver is free). RTS/CTS Mechanism:
  - **Request to Send (RTS):** Short control frame sent by transmitter requesting access
  - **Clear to Send (CTS):** Receiver replies w. CTS if ready to one node at a time.
    - ∗ **Deferring**: Nearby nodes hearing CTS for a different receiver address (RA) must defer for time specified in Duration field.
    - ∗ **Backoff (BEB)**: If RTS fails (no CTS received), sender assumes a collision and performs a Binary Exponential Backoff before retrying.
  - **Upside**: RTS collisions are faster, less wasteful than hidden terminal collisions
  - **Downside:** overhead is high for waiting for CTS when contention is low + network overhead

8. Medium Access Control (MAC) for Internet of Things

IoT MAC has different goals from generic wireless MAC: energy conservation as primary objective, can tolerate low throughput and higher latency. Hence, implication that protocols for WiFi (CSMA/CA) waste a lot of energy, and IoT has purpose built MAC protocols to minimize radio-on time.

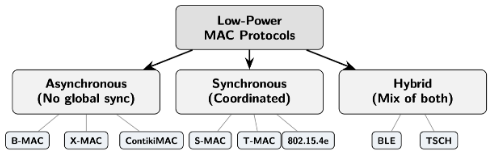
- **MAC Goals for Wireless** (E.g. WiFi): General wireless MAC protocols optimize for:
  - Low Latency, High Channel Utilization, Fairness (equal chance to communicate)
- **MAC Goals for Low-Power IoT**: Optimize for unique requirements:
  - Extremely low power consumption, low, intermittent channel utilization
  - Energy conservation becomes the primary design goal, MAC protocols must minimize power usage to prolong battery life, tradeoff latency or reduced throughput.



- **Idle listening** consumes significant energy, especially at low load, can consume between $50-100\%$ of energy needed for actual receiving. But need to listen for incoming transmissions, so cannot simply turn off radio.
- **Duty cycled MAC protocol**: periodically wake up briefly to sense transmissions/channel activity, return to sleep mode. Nodes sleep when inactive, minimize idle listening time.
- **Challenges**: Determining the optimal wake-up schedule, coordination to ensure nodes wake at the right time to detect transmissions.
- All low power MAC protocols are fundamentally about deciding when to turn the radio on and off. Trade-off latency, energy.
- **Duty Cycle Formula**: Duty Cycle $= \left(\frac{t_{\text{active}}}{t_{\text{active}}+t_{\text{sleep}}}\right) \times 100\%$

Design Space of Low-Power MAC

- **Synchronous vs Asynchronous**: Does network share common schedule?
- **Sender-initiated vs. Receiver-initiated**: who starts handshake?
- **Preamble-based vs. Schedule based**: How does sender reach sleeping receiver?



Comparison of Low-Power MAC

	B-MAC	X-MAC	ContikiMAC	S-MAC	TSCH
Type	Async	Async	Async	Sync	Sync+FH
Initiation	Sender	Sender	Sender	Both	Schedule
Preamble	Long	Short probes	Data repeats	None	None
Addressing	No	Yes	Yes	Yes	Yes
Multi-hop	Yes	Yes	Yes	Yes	Yes
Freq. hopping	No	No	No	No	Yes
Latency	Medium	Low	Low	Medium	Deterministic
Energy (sender)	High	Medium	Low	Low	Low
Energy (receiver)	Medium	Low	Low	Low	Low
Non-target waste	High	Low	Low	Medium	None
Complexity	Very low	Low	Medium	Medium	High
Reliability	Low	Medium	Medium	Medium	High
Best for	Simple WSN	General IoT	Contiki apps	Dense WSN	Industrial

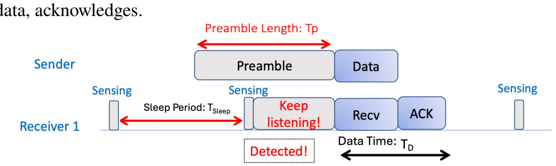
- **Selection Guidelines**:
  - Simple deploy: X-MAC/ContikiMAC. Dense w. regular traffic: S-MAC or T-MAC.
  - Industrial reliability: TSCH. Intermittent: receiver-initiated or wake-up receiver.
- **Design Principles**: Sleep by default, minimize idle listening (duty-cycle), trade latency for energy, reduce control overhead (RTS, CTS, becaons), avoid overhearing, adapt to traffic, exploit link quality, consider whole system (energy budget)
- **Information reduces energy**: Every improvement in low-power MAC is to give nodes more info, make better decisions when to sleep and wake.

8.1 Asynchronous MAC (Sender-Initiated / LPL)

Nodes operate on independent schedules. The burden of 'finding' receiver on sender.

Theory: ALOHA with Preamble Sampling

- **Steps**: Sender transmits long preamble to "wake up" receiver. Receiver periodically wakes briefly to sense for preambles. Once preamble detected, receiver stays awake, receives the actual data, acknowledges.



- **Detection**: preamble duration ($T_p \geq T_S$) must be greater or equal to receiver's sleep interval. Else, receiver may miss preamble, sleep before data starts.
- **Advantages:** Reduces idle listening significantly, saving power, simple to implement async
- **Disadvantages:** Preamble wastes energy at sender and slightly increases latency

B-MAC: Asynchronous Medium Access Control

Berkeley MAC protocol, use preamble sampling. Improvement over ALOHA preamble: Adds **Clear Channel Assessment (CCA)** and a configurable interface for duty cycling.

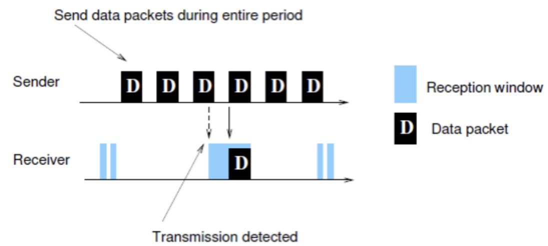
- **Preamble sampling concept**: Nodes independently choose wake-up intervals (asynchronous).
- Operational Steps:
  1. Sender transmits a long preamble
  2. Receivers periodically sample the channel
  3. If a preamble is detected, the receiver stays awake for data transmission
- **Channel sampling interval**: Determines energy usage (Receiver waking rarely saves energy, but need longer preamble, sender wastes energy, higher latency, channel blocked longer). Long preamble as duration must match sampling interval to ensure detection.
- Pros: Energy-efficient, simple, asynchronous operation
- Cons: Long preamble introduces latency and unnecessary overhead

X-MAC: Explicit MAC (B-MAC improvements)

- Replace long continuous preamble w. **short probe packets with gaps**:
  1. Sender transmits short probe packets containing the destination address
  2. Between probes, sender listens briefly for response
  3. Target receiver wakes up, checks address, sends early ACK
  4. Sender immediately transmits the data packet
- **Early ACK**: sender stops probing as soon as target responds. On average, half preamble duration. Also **reduced latency**, data transfer begins on response, no wait for full preamble.
- **Non-target receivers**: sleep immediately after read address. In B-MAC all neighbours must listen to all preambles and data.
- **Gains**: X-MAC achieves 2-3x energy savings over B-MAC by reduce preamble by 50% (on average target wakes up halfway) and eliminate non-target overhearing.
- **Limitation**: X-MAC still uses async wake-ups, sender must probe for up to entire preamble duration in worst case. Synchronous approaches (S-MAC) avoid this.

ContikiMAC

- **ContikiMAC** treats data as preamble, takes X-MAC further.
- No separate preamble or probe packets. Sender repeatedly transmits actual data packet until ACK is received. Data serves as wake-up signal. Receiver detects data, decodes, and ACKs.
- **Phase-Lock optimization**: After first successful exchange, sender learns when the receiver wakes up. Subsequent transmissions are timed to coincide with the receiver wake-up. Reduces preamble to near-zero for known neighbors.
- No wasted probe packets, phase lock eliminates preable overhead for known neighbours, energy efficient in practice.

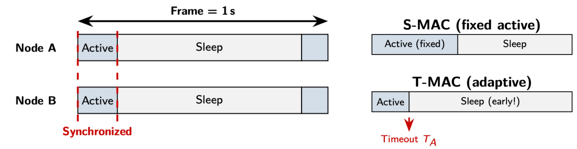


8.2 Synchronous MAC (Scheduled)

Nodes agree on a common wake-up schedule. Neighbors follow same schedule.

S-MAC: Synchronous Duty-Cycled MAC

- Nodes agree to wake up simultaneously at predetermined intervals
- Each interval split into active (listen period) and inactive (sleep period)
  1. 10% duty cycle: 10% of the time nodes listen; 90% sleep.
  2. For example: nodes listen for 100 ms, then sleep for 900 ms.
- **Advantages:** Predictable and coordinated, easier to manage communication during listening, lower sender energy than B-MAC.
- **Disadvantages:** Requires synchronization overhead (clock drift), less flexible compared to asynchronous schemes (X-MAC, B-MAC). Latency for event directly after active period.



T-MAC: Timeout - Improve S-MAC Efficiency

- **S-MAC Limitation**: Fixed active period length. Nodes waste energy on idle channel period.
- **T-MAC Improvement**: Use **adaptive active period**. Nodes stay awake only for activity.
- **Mechanism (Timeout T_A)**: If no start-of-frame is detected within T_A, node sleeps early.
- **Gains**: Under 10% load, T-MAC can save up to **5**$\times$ more energy than S-MAC.
- **Trade-off (Early Sleep Problem)**: A node might sleep before a neighbor has a chance to transmit to it. This is partially mitigated by using *Future Request-to-Send (FRTS)*.

TSCH: Time-Slotted Channel Hopping (802.15.4e)

Standard: Part of IEEE 802.15.4e. Used in industrial IoT (WirelessHART, 6TiSCH).

- **Concept**: Combine **TDMA** (time slots) w. **Frequency Hopping** (packets change channels).
- **Slotframe**: Time is divided into repeating slotframes. Each slot (typically 10ms) provides enough time for TX + RX + ACK + Guard time.
- **The Cell**: A communication link is assigned a specific (Time Slot, Channel Offset).
- **Key Advantages**:**Reliability**: Channel hopping avoids persistent interference (e.g., Wi-Fi on specific freq), **Determinism**: Guaranteed latency bounds as no contention within assigned cells, **Scalability**: Multiple pairs can communicate in same time slot using different channels.

8.3 Other MAC protocols

Wake-Up Receivers (WuRx)

- **Add a second, ultra-low-power receiver**: listens continuously while **Main Radio** stays in deep sleep (5–20mA vs. 1–100μA).
- **Operation**: The WuRx detects a specific wake-up signal and triggers an interrupt to 'power on' the main radio for data transfer.
- **Advantages**: Eliminates *idle listening* energy on the main radio. Near-zero power consumption with almost zero latency (no waiting for duty cycle wake-up).
- **Challenges**: **Low sensitivity** (~ -55dBm) means much shorter range than main radio (~ -100dBm), **False Wake-ups**: Interference can trigger main radio, waste power.

Sender-Initiated vs. Receiver-Initiated MAC (who triggers handshake)

- **Sender-Initiated**: Sender transmits preambles/probes until the receiver wakes up. (E.g. B-MAC, X-MAC, ContikiMAC.) **Trade-off**: Simple logic, but long preambles waste sender energy and increase collision probability.
- **Receiver-Initiated**: Receiver wakes up and broadcasts a short **beacon** ('I'm awake, send data now'). Senders listen for this beacon before transmitting. (E.g. RI-MAC, A-MAC.). **Tradeoffs**: Eliminate long preambles; more efficient for multiple senders. But, sender must stay awake (idle listen) to catch the beacon; beacons can collide.

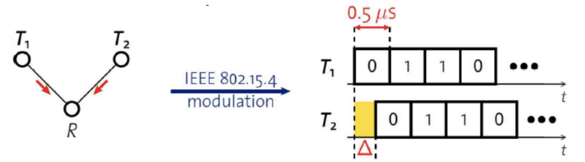
8.4 Beyond Collision Avoidance: Exploiting Interference

Capture Effect (Decoding)

- **Definition**: The ability of a receiver to successfully decode the **strongest** signal during a collision, effectively 'ignoring' weaker interferers.
- **The SIR Rule**: Typically requires a Signal-to-Interference Ratio (SIR) > 3–6dB.
- **Quantitative Evidence**: A signal at -70.89dBm (strong) can achieve 99% reception despite collisions if it has a $\sim$10dB advantage over the weak signal.
- **Impact**: Provides a natural *spatial priority* for closer nodes and increases actual ALOHA throughput beyond traditional theoretical limits (Ge^{-2G}).

Synchronous Transmissions (Time-based)

- **Concept**: Multiple nodes transmit the **exact same packet** at the **exact same time**.
- **The 500ns Rule ($\triangle$)**: R can receive packet with high probability if $\triangle$ is small. For IEEE 802.15.4, if timing offset between signals is $\Delta < 500$ns (approx. 1/32 of a symbol duration), they arrive in phase.
- **Constructive Interference**: Instead of collision corruption, the signals reinforce each other.
- **Benefit**: More transmitters: higher signal energy, higher probability of reception at receiver.



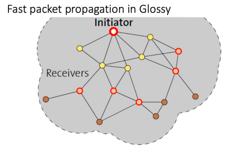
Network Layer Application: Glossy Flooding

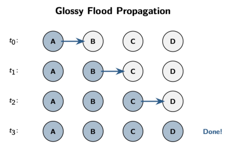
- **Traditional Simple Flooding Mechanism**: Nodes rebroadcast received packet once with duplicate detection.
- **Challenges**: **Broadcast Storm**: $O(n)$ transmissions, massive redundancy and energy waste. **Collision Cascade**: Nodes at same hop distance rebroadcast at similar times, making collisions nearly certain without random delays. **Latency**: Accumulates due to random backoffs (jitter). A 10-hop flood can exceed 150ms.

Solution: Glossy (Sync-TX Network Flooding)

Paradigm Shift: Stop avoiding collisions; exploit them as a beneficial feature.

- **Mechanism**: Receivers use hardware interrupts to retransmit a packet *immediately* upon reception to meet the < 500ns synchronization requirement.
- **Performance**: **Speed**: Floods a 200-node network in $\sim$5ms (vs. 100+ ms for traditional flooding). **Reliability**: high delivery rate from deterministic constructive interference.
- **Applications**: Sub-microsec time sync, firmware updates, **Low-Power Wireless Bus (LWB)**.





8.5 Li-Fi Light Fidelity (802.11bb)

- **Definition**: Wireless communication using visible light (VLC) instead of radio frequencies. Standardized as **IEEE 802.11bb**.
- **Transmitter**: Modulate LED intensity at speeds imperceptible to the human eye. **Receiver**: A photodetector or camera decodes the light pulses back into digital data.
- **Dual Use**: Same LED infrastructure provides room illumination and data transmission.
- **Advantages**: **Massive Bandwidth**: Entire visible light spectrum ($\sim$300–800THz), orders of magnitude wider than entire RF spectrum. **Security**: Light does not penetrate walls, no eavesdropping. **RF-Safe**: No electromagnetic interference, good for hospitals, aircraft, underwater environments where RF attenuates quickly. **License-Free**.
- **Disadvantages**: Require **Line-of-Sight (LoS)** and direct or near-direct alignment. **Interference**: ambient sunlight or other light sources create a high noise floor. **Uplink Difficulty**: While ceiling LEDs easily provide *downlink*, an *uplink* (device to ceiling) often requires additional IR LEDs or hybrid RF solutions.
- **Typical Applications**: **Indoor Positioning**: LED ID beacons for precise location tracking, **D2D (Device-to-Device)**: Short-range data transfer (e.g., screen-to-camera), **Hazardous Environments**: Safe for use in petrochemical plants or mines where RF sparks are a concern.

9. Wireless Localization

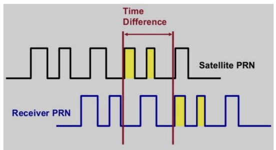
- Localization systems determine the physical coordinates of an electronic device such as IoT nodes, drones, mobile phones or a subject. **Types of Target Physical Coordinates**:
  - **Global (absolute) coordinates:** defined using an external reference system (e.g., GPS), meaning coordinates are absolute and universally understood.
    - ∗ Global coordinates: (37.7749° N, 122.4194° W)
  - **Relative coordinates** defined with respect to another coordinate system, using transformation like rotation, reflection, or translation
    - ∗ E.g., locating an AirTag indoors, for example, inside a room

9.1 GPS — Global Positioning System

- Satellites placed in Medium Earth Orbit (Altitude 20,000 KM)
- Orbit twice a day, $\geq$ 4 satellites visible from anywhere on Earth at all times
- Steps of GPS positioning:
  1. GPS satellites broadcast their positions and time
     - Position: Global coordinates of the satellite
     - Time: A precise timestamp of when the signal was sent (Time of Transmission)
  2. The GPS receiver (e.g., phone) receives signals from multiple satellites
     - It logs the exact timestamp (Time of Arrival) when each signal arrives
  3. The GPS receiver calculates the distance from each satellite
     - Time of flight = Time of Arrival - Time of Transmission
     - Distance = time of flight $\times$ speed of light (3.0×10^8 m/s)
  4. Using trilateration, your device computes your exact location
     - With dist. measurements from each beacon, we determine the position
     - Goal: find intersection point of circles(2D) / spheres (3D)
- GPS requires at least 4 satellites
  - 3D scenario: with 3 satellites, trilateration results in two possible locations. If target device on ground, the incorrect point can be easily eliminated.
  - ***However***: Not all targets are on the ground, e.g., aircraft. Additionally, may be clock (time) drift in GPS receivers, e.g., mobile phone

GPS receiver (e.g. phone): Determining distance

- Each satellite periodically sends a unique pseudo-random no. sequence
  - Receivers also locally generate the same sequence in synchronized fashion
- Receivers measure time of arrival of the sequence, use offset in sequence to calculate Time of Flight and distance.
- Transmission also includes time of transmission of sequence, location of satellite at time



Distance = time of flight X speed of light

- Satellites have very accurate (expensive) clocks, and all satellites are precisely synchronized.
- Receivers have less accurate clock, e.g., mobile phone, much cheaper. The 4th satellite helps solve for the receiver's clock error.
- Overall, solves for 4 unknowns: `latitude, longitude, altitude, clock drift`

Physical layer of GPS

- Frequency: 1.2GHz and 1.5GHz, $10 - 15$MHz. BPSK modulation.
- Receiver needs to know additional info: Current time, position of each satellite
- GPS transmission has this data layered on top (50bps)
  - **Ephemeris**: valid for up to 4 hours. Listen (up to) 30 seconds gets time, satellite position.
  - **Almanac**, valid for up to two weeks. Listen for 12.5 minutes gets all satellites positions.

Assisted GPS

- Cell phone GPS is so quick by downloading **almanac** from the internet
- Bootstraps location information, cell towers give coarse position, enables devices to know which satellites are overhead

Beacon Nodes (e.g., GPS satellites)

- Necessary to localize a network in a global coordinate system, also they can be used for finding relative coordinates. At a minimum:
  - Three (non-collinear) beacons required to define a global coordinate system in 2D.
  - Four non-coplanar beacons for 3-D coordinates
- Beacon Placement: Has significant impact on localization
  - If the beacons are spread out around the target area instead of being clustered together, we get more accurate positioning
  - Adding an extra beacon somewhere centrally can also improve results

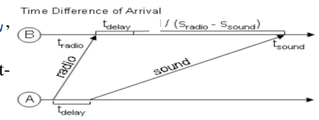
9.2 Ways to measure distances

Time of Flight (ToF) / Time of Arrival (ToA)

- **Mechanism**: Determine distance by knowing exact position of infrastructure, Transmit time, Receive time, Signal velocity (e.g. light)
- Infrastructure transmits (can happen constantly), device can listen only when needed.
- **Conditions**: Requires time synchronization between infrastructure and device (Synchronization must be good: 1μ s $= 300$ meters). Additional considerations: Speed of signal propagation, Distance to be measured, Clock accuracy
- E.g.: Speed of light (c) $\approx 3 \times 10^8$ m/s, Distance travel in 1ms = 300m, Clock accuracy needed for meter accuracy ($1/c$) ≈ 3ns

Time difference of arrival — (TDoA)

- (Target) Device transmits and infrastructure (beacon) listens transmission. Multiple connected infrastructure nodes receive at different times based on distance.
- **Mechanism**: Determine distance by knowing exact position of infrastructure, Time of arrival at two different locations, Signal velocity (i.e. speed of light).
- Does not require synchronization with infrastructure!
  - Still requires synchronization between infrastructure nodes
  - Does require device to transmit 'loud' enough for infrastructure to hear it
- With different mediums:
  - TDoA: $\frac{d}{S_{sound}} - \frac{d}{S_{Radio}} + t_{delay}$, Works well in LOS environment
  - Requires additional hardware but works better with calibration



Ultrasound

- **Advantages**: Solves barrier problem (cannot pass through solid objects like walls). Human spaces already designed to contain sound. Easier to get high-accuracy: Sound much slower than light. Less synchronization needed for same accuracy.
- **Disadvantages**: More energy to transmit, Slower update rate (still sub-second), Limited range, Pets can hear it

Received Signal Strength Indicators (RSSI)

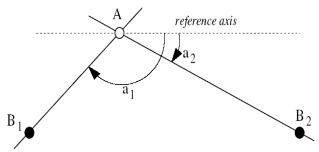
- RSSI measures signal strength to estimate distance. In theory, RSSI varies with $1/d^{\alpha}$
  - α depends on the wireless environment (typically 2 or 3)
  - Common assumption: closer nodes receive stronger RSSI than those further away
- In practice, RSSI range measurements contain noise, several meters off, highly non-uniform
  - Radio propagation differs over different surface (water, asphalt etc.), obstacles (wall, furniture etc.). **Problem:** Pathloss is NOT only due to distance

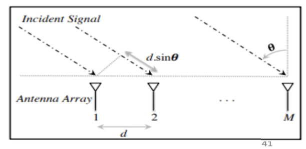
Fingerprinting using RSSI

- Shifts from trilateration to mapping: Use RSSI patterns of known locations.
- Phase 1 - Setup: Measure signal strength from multiple Access Points at various locations. Record measurement in a database - mapping signal strengths to specific locations.
- Phase 2 - Localization: Device measures real-time signal strength from surrounding APs. Compare readings to the database to determine the most likely location.
- **Wi-Fi Fingerprinting**: Repurpose existing infrastructure for localization (e.g. Wifi points). Enable localization with unmodified hardware, no extra infra. (E.g. WiFi RADAR)
- **Challenges**: Efforts to create database: manual measurements at locations, Non-static environment: signal strength changes with obstacles, periodically re-measure and update database, Measurements might vary between devices (how you hold, antennas of devices etc).
- **Variations:** Measurements can use several access points simultaneously to improve accuracy. No need Wi-Fi based, can use cellular networks or deploy own BLE beacons
- **Accuracy:** State-of-the-art: median accuracy of 0.5-1.5 meters. (Enough to pinpoint room etc.) Barrier problems depend on walls.

Angle of arrival (AoA)

- Measures direction of incoming signal using antenna array
- Using 2 anchors (B1, B2), A can determine position. BLE 5.1 includes AoA localization
- Leverage antenna arrays, measure differences in arrival time ($\triangle t$) across antenna elements
- d (antenna spacing), $\triangle t \times$ speed of light $= d \sin\theta$; $\implies \sin\theta =$ speed of light $\times \frac{\triangle t}{d}$





10. Wireless Sensor Networks

10.1 Neighbour Discovery

Once deployed, sensors have to initiate neighbour discovery to connect with one another.

- **Naive approach**: Continously listens for neighbour to transmit signals until hears transmission. To wake up and constantly listen, battery drained **quickly**. Not energy efficient.

Synchronous Neighbour Discovery

- **All nodes synchronized** to same clock/time. Nodes scheduled to wake up periodically at same time to transmit/receive.
- deterministic approach — provide guarantee bounds on discovery latency
- **Disadvantages**: Synchronization overhead, consumes significant power. Slow discovery process to detect all nodes. (collisions) Scalability issues as network grows, more nodes to sync.

Asynchronous Neighbour Discovery

- **No clock synchronization needed**: Each device uses own clock to divide timeline into time slots. Independently wakeup for TX/RX at random time slots.
- No restrict wake-up schedules $\rightarrow$ much lower average discovery latency

Birthday Protocol - Asynchronous neighbor discovery

Easy to implement + Good average discovery latency but unbounded worst case

1. Nodes have repeating periods. E.g. each period has 10 time slots.
2. Node randomly pick wake-up slots (shaded) (e.g., 1 slot per period (10% duty cycle))

Model probability of birthday paradox

- Two nodes randomly choose to wakeup k time slots (shaded boxes) out of a total n slots (periods). What is the probability that they wakeup at the same time slot at least once?
- Probability (Q) can be written down as:
  - (Duty-cycle: 1%) $n = 100$, $k = 1$, $Q(100, 1) = 0.01$
  - (Duty-cycle: 5%) $n = 100$, $k = 5$, $Q(100, 5) = 0.23$
  - (Duty-cycle: 1%) $n = 1000$, $k = 10$, $Q(1000, 10) = 0.096$
  - (Duty-cycle: 5%) $n = 1000$, $k = 50$, $Q(1000, 50) = 0.928$
- $Q(n,k) = 1 - \dfrac{\binom{n-k}{k}}{\binom{n}{k}}$
- Drawbacks: probabilistic bound for discovery time, some nodes may take very long

DISCO - Asynchronous deterministic neighbor discovery

- Scheduling wake-up timeslots at multiples of prime numbers to ensure deterministic discovery. Nodes can have different duty cycles.
  - Each node picks two prime numbers p_{i1} and p_{i2}
  - Node **wakes up every** p_{i1} **and** p_{i2} **time slots** to $\uparrow$ discovery chances
  - The duty cycle is $(1/p_{i1} + 1/p_{i2})$
  - If both nodes pick the same pair of prime numbers:
    - Duty cycle $\approx (p_{i1} + p_{i2})/p_{i1}p_{i2}$, Worst Case time $\approx p_{i1}p_{i2}$

10.2 Routing

If all N nodes connected directly, $N(N-1)$ or $O(N^2)$ uni-directional links

Structure of wired networks

- Hierarchical and structured: *switches*, *routers* for managing connections and route data.
  - Switch connects multiple devices within a local network (e.g., at home or in an office)
  - Router connects different networks together (e.g., your home network to the Internet)
- Add more switches/routers reduces # of direct links needed and allows network scaling.

Structure of wireless networks

- Wireless network are typically single-hop: All stations/devices communicate directly with base stations or (Access Points) (E.g. WiFi, Cellular Network, LoRa)
- **Multi-hop**
  - Extends network reach/coverage for the same number of base stations
  - Provides more routes, potentially more robust

10.3 Routing Metrics: Factors to consider in Routing Decisions

- Energy efficiency
- Reliability: ensure packets successfully delivered (e.g., ETX)
- Hop Count
- Delay: how long it takes for data to reach the destination
- Other factors: network congestion / throughput

Energy Consumption

- Free space model: $P_r = CP_t/d^\alpha$
  - C is a constant — α is an exponent, typically 2 or 3
- Transmission over a total distance d, minimum received power P_0
- **Single-hop** transmission to cover distance d, calculate transmission power (P_1):
  - $P_0 = CP_1/d^\alpha$ or $P_1 = P_0/C * d^\alpha$
- **Multiple-hop** (N) transmissions to cover distance d, calc transmission power (P_N) per hop:
  - $P_0 = CP_N/(d/N)^\alpha$ or $P_N = P_0/C * (d/N)^\alpha$
  - Total transmission power is $NP_N = NP_0/C * (d/N)^\alpha$

Reliability

- Ensure packets successfully delivered, minimize packet loss
- Routing Metric: "expected number of transmissions" to successfully transmit a packet (ETX)
- Measure link quality over time to determine each link's reliability
  - Link quality varies between nodes
  - Fewest hops might not be the "most reliable" path

Measure Packet Reception Ratio (PRR)

- A transmitter sends S packets over a period of T seconds. The receiver records number of packets successfully received, denoted as R. $PRR = R/S$
- **Overheads of Measuring PRR**: Trade-off communication cost and reaction time.
  - Assume ZigBee radio is used (250Kbps), each transmission takes 1ms ($\sim$250bits, $\sim$30bytes) and link measurement probes should not collide
  - If T=1s, S=100, Measurement takes up 10% of bandwidth 1 node. If there are 10 nodes, 100 probes need to be transmitted in 1 sec (100% of bandwidth used only for measurement!)
  - If T=100s, S=10, Measurement takes up very little bandwidth, but it takes a long time to get an accurate estimate of link quality. If link quality changes during that time, the measurement may be inaccurate. Good S and T values application dependent.

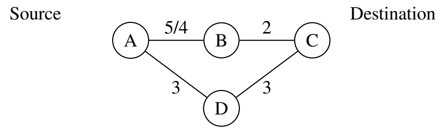
Calculate Expected number of Transmissions: ETX

- ETX (expected transmissions) is a commonly used routing metric
- Using Packet reception ratio (PRR): $ETX = 1/PRR$

10.4 Routing Algorithms

Route using shortest path algorithm

- Many practical routing algorithms based on notion of shortest path between two nodes. Each link assigned a positive number called its length. Route each packet along minimum length (or shortest) path between source and destination.
- If length is always 1, shortest path becomes minimum hop routing
- By defining the length using metrics like ETX, energy, delay etc, other paths can be obtained



Classification of Routing Protocols

- **Proactive (Routing Table based):** when a packet needs to be forwarded, the best route is already known (stored in the routing table)
- **Reactive (Demand-based):** Determine the best route only when there is data to send

Proactive Routing

- Periodically query for the possible routes in the network. Save all routes that are important (or all routes), also determine routes whenever topology changes (nodes join/leave)
- **Upside:** when a packet arrives, route to destination is already known
- **Downside:** requires more memory and effort on part of routers. Wastes some network bandwidth on checking for route changes

Reactive Routing

- Opportunistically build up a map of the routes through network, hopefully optimal routes.
- Map routes in reaction to a packet arrival.
- **Rationale**: Sensor devices are slow and limited, most likely to resend to same prior address. Discover a route when it is needed, then cache for next time

Ad-hoc Networks

- **A Mobile Ad hoc Network (MANET)** is an autonomous system of mobile (non-static) nodes (MS) (also serving as routers) connected by wireless links. No infrastructure exists.
- Network's wireless topology may change dynamically in an unpredictable manner as nodes are free to move, each node has limited transmitting power.
- **Assumption**: Due to node mobility, routing path may be available, needs to be "discovered"

10.5 Ad-hoc On-demand Distance Vector Routing (AODV)

- **Ad-hoc:** dynamic and temporary networks
- **On-Demand**: Routes are only created when a source node has data to send.
- **Distance vector:** routing metric is the number of hops a packet has to pass (Hop Count)

AODV Routing Table

Each node maintains a table that stores:

- **Destination IP Address**: The final target.
- **Next Hop**: Stores only the immediate neighbor towards the destination, not the full path.
- **Hop Count**: Distance to the destination.
- **Destination Sequence Number**: An indicator of route **freshness**.
- **Lifetime**: Expiration time of the route.

The 2-Step Route Discovery Process

1. Route Request (RREQ) – Discovery (Broadcast)

- **Action**: When route isn't cached, source **floods** an RREQ (nodes forward it, incrementing hop count), till reaches destination or node with valid route to destination.
- **Reverse Path**: Each intermediate node records which neighbor sent the RREQ, creating a path back to the source.
- **AODV Route Requests (RREQ) Packet Structure**:

Request ID	Source Address	Destination Address	Source SeqNo	Destination SeqNo	Hop Count

- **Sequence Numbers**: Monotonically increasing values used to determine if information is "fresh." A node only updates its table if the incoming SeqNo is higher than its stored value.
  - **Source SeqNo**: E.g. like a broadcast number, used by receivers to learn the 'freshness' of the path *back to the source*.
  - **Destination SeqNo**: Used by the source to specify how 'fresh' the route *to the destination* must be that it will accept. If source does not know destination's SeqNo, it can set it to zero, meaning it will accept any route.
  - **The Rule of Freshness**: A node updates its routing table only if:
    1. Incoming SeqNo $>$ Stored SeqNo (Newer information), or (Shorter path):
    2. Incoming SeqNo $=$ Stored SeqNo **AND** Incoming Hop Count $<$ Stored Hop Count
- **Request ID**: Unique ID for RREQ to detect and discard duplicate broadcasts.
- **Hop Count**: The number of hops request has taken. Starts at 1, incremented by nodes.

2. Route Response (RREP) – Establishment (Unicast)

- **Action**: The destination (or a node with a valid route) replies with a unicast RREP.
- **AODV Route Response (RREP) Packet Structure**:

Source Addr	Destination Addr	Destination SeqNo	Hop Count

- **Forward Path**: The RREP follows the reverse path back to the source, establishing the forward route for data in the routing tables. Includes destination sequence number as a sense of recency. Note dest. addr. still refers to the original target node, same as the RREQ.
- **Logic Check**: The **Destination Addr** and **Dest SeqNo** still refer to the **Target Node**. The RREP updates the Dest. SeqNo before sending RREP to prove required recency.
- **Generating the RREP**:
  - **By Destination Node**: Increments its sequence number to ensure it is strictly greater than the one in the RREQ. This forces the network to accept the new route.
  - **By Intermediate Node**: Can only reply if it has a route with a $DestSeqNo \geq$ the one in the RREQ. It *cannot* increment the value; it simply relays the cached $DestSeqNo$.

Route Maintenance (Self-Healing)

- **Link Break**: Detected when a neighbor fails to acknowledge a transmission.
- **RERR (Route Error)**: Node that detects break sends a RERR to all 'upstream' neighbors.
- **Recovery**: Source node receives the RERR, deletes the stale route, and starts a new RREQ if it still needs to send data.

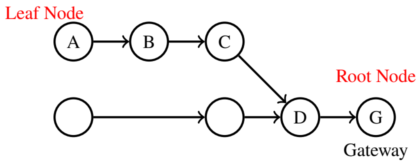
Tradeoffs for Reactive Routing

- **Upside: Efficiency**
  - No transmissions unless demand: Save battery, bandwidth, avoid periodic 'heartbeats'.
  - **Dynamic Tolerance**: Routes can break/change w/o penalty if no data currently being sent.
- **Downside: Performance Costs**
  - Large, Variable Delay: The first packet must wait for the full RREQ/RREP 'handshake' to complete across the network before it can be sent.
  - **Discovery Overhead**: In high-mobility or high-traffic networks, frequent RREQ broadcasts can lead to a 'Broadcast Storm,' clogging the channel.
  - **Stale Cache Issues**: If a cached route is broken, the sender won't know until the data fails and a **RERR** is received, forcing a costly discovery restart.

10.6 Load Balancing in Wireless Networks

Many-to-one Sensor Networks

- Data collection follows a spanning tree
  - All data is collected on a single node (gateway)
- Routing is simple:
  - Each device only needs to remember hop to gateway
  - If gateway wants to send message back, it must include a full hop path
- **Power consumption for each node**
  - Per-node power consumption for data transmission and reception grows exponentially from the leaves to the root of the tree
  - Power consumption increases at least linearly when nodes are closer to the sink
  - Typical case is much worse
  - Consequence: the power sources of the nodes close to the sink (gateway) deplete faster.



In-network data aggregation

- To mitigate cost of forwarding, compute relevant statistics along the way: mean, max, min, median etc. (e.g. Node C sends mean of (A, B, C), instead of three separate messages).
- Forwarding nodes aggregate the data they receive with their own and send one message instead of relaying multiple messages. Prevent exponential growth of messages.
- **Issues:**
  - Location-based information is lost
  - Distributed computation of statistics: For mean, nodes needs to know both the mean values and the sizes of samples to aggregate correctly, and for median, only an approximated computation is possible
- **Applications:**
  - Especially useful in a query-based data collection system
  - Queries regard a known subset of nodes
  - Aggregation function can be specified

11.0 Evolution of WiFi

In 20 years, WiFi has evolved from 2 Mbps to 9.6 Gbps. Achieved through:

1. **Capable Modulation**: Shifting from Spread Spectrum to OFDM/OFDMA.
2. **Bandwidth Expansion**: From 2.4 GHz to 5 GHz/6 GHz, wider channels (20 to 160 MHz).

	Protocol	Year	Frequency	PHY	Max Rate	Range
-	802.11	1997	2.4 GHz	FHSS/DSSS	2 Mbps	20 m
1	802.11b	1999	2.4 GHz	DSSS	11 Mbps	35 m
2	802.11a	1999	5 GHz	OFDM	54 Mbps	35 m
3	802.11g	2003	2.4 GHz	OFDM	54 Mbps	38 m
4	802.11n	2009	2.4/5 GHz	OFDM + MIMO	600 Mbps	70 m
5	802.11ac	2013	5 GHz	OFDM + MU-MIMO (downlink only)	3.4 Gbps	35 m
6	802.11ax	2021	2.4/5/6 GHz	OFDMA + MU-MIMO	9.6 Gbps	35 m

11. WiFi (802.11)

- **Purpose**: High-speed, short-range (10–30m) indoor communication.
- **History**: FCC opened ISM bands (1985); Wi-Fi Alliance (1999), certify compatibility.

WiFi Network Architecture

- **Star Topology**: All client devices connect to a central **Access Point (AP)**.
- **Basic Service Set (BSS)**: The combination of one AP and its connected clients.
- **SSID (Service Set ID)**: The network name, broadcast by the AP in **Beacons** so clients can find, identify and join the network.

11.1 MAC Layer: How WiFi Avoids Collisions

WiFi uses **CSMA/CA** (Carrier Sense Multiple Access with Collision Avoidance) as wireless radios cannot usually detect collisions while transmitting. **CSMA/CA Process:**

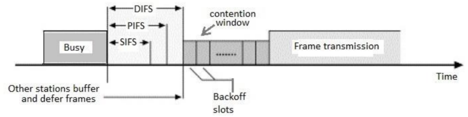
1. **Carrier Sense**: Listen to the medium.
2. **If Free**: Wait for a specific **IFS** (Inter-Frame Spacing). If still free, **Transmit**.
3. **If Busy (Random Backoff)**: Select a random number of **Backoff Slots**.
   - **Count Down**: Only decrement slots while the medium is idle.
   - **On Expiry**: If the medium is busy when the counter reaches zero, **increase** the maximum backoff range (Binary Exponential Backoff) and repeat.

Slot Time: basic unit of time for protocol:
$\text{Slot Time} = \text{Rx-to-Tx switch time} + \text{Processing delay} + \text{Propagation delay}$

11.2 Traffic Priority - Tiered Contention Multiple Access (TCMA)

WiFi uses different IFS (inter-frame spacing) durations to ensure certain traffic classes / packets (like ACKs) can 'cut the line' before new data starts. (Prioritize communication)

- **Hierarchy**: **SIFS** $<$ **PIFS** $<$ **DIFS**
- **SIFS (Short IFS)**: Used for **Acknowledgements (ACKs)**. Because SIFS is the shortest, an ACK will always be sent before any other device can start a new data transmission.
- **DIFS (Distributed IFS)**: Used for standard data packets. This is the 'normal' waiting time.
- **Logic**: Shorter wait time = Higher priority. (PIFS: Point Coordination IFS, for control frames)

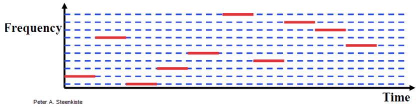


11.3 Spread Spectrum (Legacy WiFi)

The goal of Spread Spectrum is to use a **much wider bandwidth** than required for the data rate to gain robustness against interference and noise.

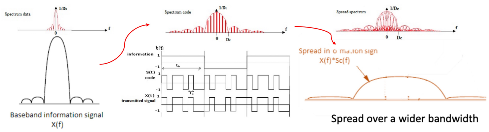
1. Frequency Hopping (FHSS)

- **Mechanism**: The transmitter 'hops' between 80 narrow channels (1 MHz each) in a pseudo-random sequence known to both sender and receiver.
- **Pros**: Simple hardware; resists interference on specific frequencies.
- **Cons**: Slow data rates due to switching delays ('settling time' of radio during a hop).



2. Direct Sequence (DSSS)

- **Mechanism**: Each data bit is multiplied by a higher-rate **Chipping Code**. This 'smears' the bit's energy across a wide frequency band (22 MHz in 802.11b) (Phase Modulation Variant).
- **Robustness**: Even if noise destroys part of the signal, the original bit can often be recovered from the remaining 'chips.' (resistant to narrowband interference). (Both RX and TX knows chipping code.) Allow multiple users to share band using different codes.
- **Advantage over FHSS**: Higher Data Rates because transmission is continuous; no time is wasted hopping between frequencies.
- Essentially, each bit is multiplied by a sequence of bits with a much higher data rate. Data Bit (t_b) $\times$ Chipping Code (t_c) = Transmitted Signal. As $t_c \ll t_b$, bandwidth increases.



Spread Spectrum Summary

- **Key Idea**: Traffic from other users appears as low-level background noise as they use different hop sequences or chipping codes. Spreading signal over wide spectrum (FHSS: multi-freq hop or DSSS: single frequency), more resilient to interference and eavesdropping.
- **Redundancy**: Extra bandwidth provides frequency-domain redundancy.
- **Legacy**: Used in 802.11 (Original), 802.11b; foundation for GPS, 3G (CDMA) cellular. Each pair of TX and RX must coordinate so RX can apply same code to decode data.

802.11b Channels (2.4 GHz)

- **Width**: 22 MHz per channel (14 total), **Spacing**: 5 MHz spacing between channel centers.
- **Overlap**: Most overlap significantly, only **Channels 1, 6, and 11** considered non-overlapping.

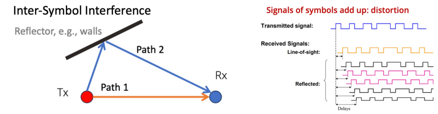
11.4 OFDM: Orthogonal Frequency Division Multiplexing

The Problem: Multipath Interference: High-speed wireless communication faces two physical challenges due to signals reflecting off walls and objects, creating multiple paths.

- **Frequency Selective Fading**: Certain frequencies in a wide band get 'cancelled out' by reflections while others pass through. Significant at high data rates and on wide-band channels.
- **Inter-Symbol Interference (ISI)**: At high data rates, bits sent so quickly (short symbol duration), reflected 'echo' of previous bit crashes into the current bit, causing distortion.

Solution: Multi-Carrier Modulation (FDM): Instead of sending one very fast stream of data on a single carrier, we distribute bits over **N subcarriers** in parallel.

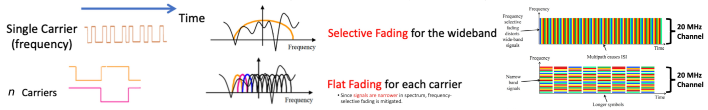
- **Subcarriers**: Each is a narrow-band signal with its own center frequency.
- **Symbol Duration**: Since each subcarrier only handles $1/N$ of data, can send N times **slower**.
- **Result**: Longer symbol durations make the signal much more resistant to ISI.



Mechanism 1: Multi-Carrier Modulation (FDM)

The core idea is to distribute bits over N narrow subcarriers instead of one wide one.

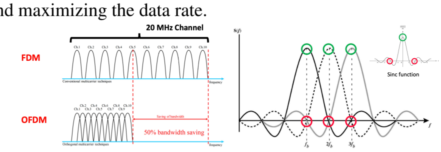
- **Symbol Duration Extension**: If you split the data across 52 subcarriers, each subcarrier only handles a fraction of the data. This allows each 'symbol' to last **N times longer** in time.
- **Why this works**: Longer symbols are 'slower.' Reflections (Multipath) finish bouncing around before next symbol starts, effectively **eliminating Inter-Symbol Interference (ISI)**.
- **Narrow-band Advantage**: Each subcarrier is so narrow that it experiences 'flat fading' (easier to fix) rather than 'selective fading' (which destroys wide signals).



Mechanism 2: The 'Orthogonal' Advantage (Achieve higher data rate)

How do we pack subcarriers densely without them bleeding into each other?

- **Traditional FDM**: Requires 'Guard Bands' (wasted empty space) between channels.
- **OFDM**: Uses **Orthogonality**. The subcarriers are mathematically spaced so that the **peak power** of one carrier happens exactly at the **null (zero) point** of all other carriers.
- **Spectral Efficiency**: This allows subcarriers to overlap significantly, removing the need for guard bands and maximizing the data rate.



Implementation & 802.11a

The math required to turn bits into these overlapping waves is complex:

- **Signal Processing**: `IFFT` (Inverse Fast Fourier Transform): Used at the Transmitter to combine all subcarriers into one signal. `FFT` (Fast Fourier Transform): Used at the Receiver to split the signal back into individual subcarriers.
- **802.11a (First WiFi with OFDM) Specs**:
  - **Band**: 5 GHz (cleaner airwaves, more channels than 2.4 GHz). **Speed**: 54 Mbps.
  - **Modulation**: **QAM** (Quadrature Amplitude Modulation) to pack more bits per subcarrier.
- **OFDM vs DSSS, Trade-offs**: **Evolution:** Replace DSSS with Orthogonal Frequency Division Multiplexing. **DSSS (Legacy)** has high redundancy, low speed. One signal 'spread' thin. **OFDM (Modern)** has high complexity, throughput, spectral efficiency. Robust against narrowband interference, multipath effects.

11.5 MIMO: Multiple Input Multiple Output

Concept: Why use Multiple Antennas? For wired, add more wires for more speed. In wireless, use multiple antennas to create 'virtual wires' in the air.

- **Architecture**: N transmit antennas, M receive antennas, create $N \times M$ subchannels.
- **Spatial Multiplexing (Speed)**: Sending *different* data streams simultaneously across antennas. This provides a huge boost in throughput without needing more bandwidth.
- **Antenna Diversity (Reliability)**: Sending and receiving multiple copies of the *same* signal. If one path is blocked by a wall, another antenna likely has a clear line of sight.

Advanced Signal Processing

- **Interference Recovery**: While signals overlap in the air, the receiver uses complex math to 'unmix' the streams based on the unique spatial signature of each path.
- **Beamforming**: Instead of broadcasting in all directions, the AP uses the interaction between antennas to **focus energy** specifically toward the client.
- **Benefit**: Higher Signal-to-Noise Ratio (SNR) and better range.

802.11n (WiFi 4): Use MIMO

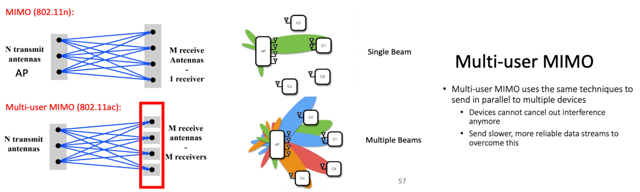
2009: 802.11n standard transformed WiFi from single-lane to multi-lane.

- **Wider Channels**: Doubled bandwidth from 20 MHz to **40 MHz**. (Channel bonding: combine two adjacent channels into one big channel)
- **Dual-Band**: The first standard to work on both **2.4 GHz** and **5 GHz** bands.
- **Throughput**: Boosted maximum rates up to **600 Mbps** (up from 54 Mbps in 802.11g).
- **Legacy**: 802.11n remains the 'baseline' for 2.4 GHz today, as many later standards (like 802.11ac) focus strictly on the 5 GHz band.

Multi-User MIMO (MU-MIMO)

Introduced with 802.11ac, solve 'crowded room' problem, many devices connected to one AP.

- **Single-User (SU-MIMO)**: AP talks one device at a time. Other devices wait (Sequential).
- **Multi-User (MU-MIMO)**: AP uses beamforming to talk to **multiple devices simultaneously** on the same frequency (Parallel).
- **802.11ac Limitation**: Supports **Downlink only** (AP to devices). Devices still contend for the medium when uploading to the AP.



802.11ac - Enhanced MIMO and wider channels

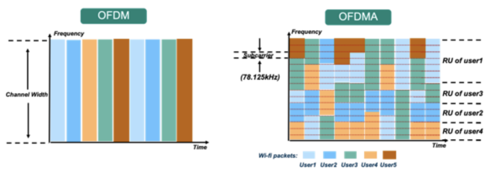
- **Implementation (2013):** Update for 5 GHz band only, supports Downlink MU-MIMO (from AP to device), channel widths up to 160 MHz, and higher modulation (256-QAM).
- **Deployment Strategy:**: Routers apply 802.11ac to 5 GHz and 802.11n to 2.4 GHz

11.6 WiFi 6 (802.11ax): OFDMA - High Efficiency Wireless

Goal: Improve total throughput across *all* devices in a crowded network (High Efficiency), rather than just the peak speed of one device.

OFDMA (Orthogonal Frequency Division Multiple Access)

- **From OFDM to OFDMA**: turn OFDM into access control mechanism. More efficient use of bandwidth in high-density environments.
  - **OFDM**: One user takes the *entire* channel for a duration of time, even if they only have a small packet to send (wasted bandwidth).
  - **OFDMA**: The channel is divided into **Resource Units (RUs)**. The AP can allocate different subcarriers to different users *simultaneously*. Multiple users can transmit simultaneously, reduced latency for small data transfers.



802.11ax (WiFi 6) - 6 GHz Band Extension and Features

- **Standard Timeline:** Approved in 2021, first devices started supporting it in 2019 (WiFi 6).
- **Spectrum**: Works in 2.4 GHz, 5 GHz, and the new **6 GHz** band (WiFi 6E).
- **6 GHz Band**: Provides 1.2 GHz of new, unlicensed bandwidth (huge increase channels).
- **Up/Down MU-MIMO**: Unlike WiFi 5, support Multi-User MIMO, both uploads/downloads.
- **1024-QAM**: Packs 10 bits per symbol (25% increase over 256-QAM) for short-range speed.

WiFi Evolution Summary

Standard	Bandwidth	Max Rate	Key Technology
802.11g	20 MHz	54 Mbps	OFDM
802.11n (WiFi 4)	40 MHz	600 Mbps	MIMO + Channel Bonding
802.11ac (WiFi 5)	160 MHz	3.46 Gbps	DL MU-MIMO + 256-QAM
802.11ax (WiFi 6)	160 MHz	9.6 Gbps	OFDMA + 1024-QAM + 6 GHz

- **Legacy (11b/g)**: Basic wireless connection.
- **Middle (11n/ac)**: Increasing the speed for a single user (MIMO, Wider Channels).
- **Modern (11ax)**: Managing high-density environments with many IoT and mobile devices (OFDMA, Scheduling).

12. Low-Power IoT

IoT devices: power, sensors, wireless communication. Categorize by operating scale (range).

Categories of IoT Wireless Networks

- **PAN (Personal Area Network)**: Range < 30 m. E.g.: **BLE**, **Zigbee**, wearables.
- **LPWAN (Low-Power Wide-Area Network)**: Range $\sim$ tens of km. Example: **LoRaWAN**. Used for agriculture and industrial monitoring.

12.1 PAN Technology: Bluetooth Low Energy (BLE)

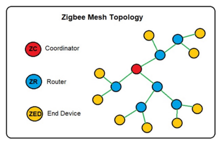
Designed to replace cables and run on coin cell batteries, use in low energy mode (LE).

- **Communication Roles**: **Advertiser**: Broadcasts packets (e.g., headphone). **Scanner**: Receives packets without connecting. **Initiator**: Attempts to form connection (becomes **Master**).
- **Channels**: 40 channels total (2 MHz wide).
  - **37, 38, 39**: Advertising channels (placed to avoid WiFi interference).
  - **0–36**: Data channels using **FHSS** (Frequency Hopping Spread Spectrum).
- **The Connection Process**
  1. **Advertising Event**: *Advertiser* broadcasts packets on advertising channels (periodic).
  2. **Initiation**: A *Scanner* receives the advertisement. If it wants to connect, it becomes an **Initiator** and sends a connection request on the same channel.
  3. **Piconet Formation**: If the Advertiser accepts, a **Piconet** (point-to-point connection) is formed. Initiator becomes **Master**. Advertiser becomes **Slave**.
  4. **Data Exchange**: Once connected, pair moves to data channels (0-36). Use **FHSS** to switch channels after connection events (typically every 4.615 ms) to combat interference, fading.
- **Types of BLE Events**: Signal wish to pair or relay broadcast message (beacons). **Periodic Advertising**: device advertises over non-primary channels using hopping pattern. (Intervals can range **7.5 ms to 81.9 s**). **Connecting**: handshake pairing device to host.

12.2 PAN Technology: ZigBee

Based on IEEE 802.15.4, operating at 2.4 GHz using DSSS.

- **Mesh Topology**: Unlike BLE (Star), Zigbee supports **many-to-many** mesh networking, allowing data to hop through routers.
- **Network Components**:
  - **ZigBee Controller (ZC)**: One per network; initiates network functions (assign logical addresses, permits node to join or leave mesh).
  - **ZigBeeRouter (ZR)**: Optional; handles multi-hop routing and joins nodes, handles some of the load of mesh network hopping and routing coordination.
  - **End Device (ZED)**: Simple sensor; no routing logic (e.g. light switch).



	BLE	ZigBee
Number of nodes	Supports few nodes	Supports many nodes
Range	10-50 m	300 m
Operating System	Android, iOS, Windows 8, OS x	Not currently compatible
Topology	Star	Mesh
Over the air data rate	125 kbit/s – 1 Mbit/s – 2 Mbit/s	250 kbit/s
Transmit power	10 mW	100 mW
Ideal use case	User traveling through a group of connected things in a defined space	Home automation

12.3 Low-Power Wide-Area Network LPWAN: LoRaWAN

Long-range Wide-Area Network LoRaWAN: most popular LPWAN protocol.

- **Physical Layer (LoRa)**: Uses **Chirp Spread Spectrum (CSS)**. (Freq. Modulation Variant)
  - **Chirp**: Signal frequency increases linearly over time. (like chirp)
  - **Spreading Factor (SF)**: Controls symbol duration. Spreads signal across wide frequency band. Higher SF = longer range, robust to noise, multipath fading but **lower data rate**.
  - **Encoding**: n bits encoded by dividing bandwidth into 2^n possible initial frequencies.
  - **Orthogonal**: Diff SFs do not interfere, simultaneous transmissions on same frequency.
- **MAC Layer**:
  - **Star Topology**: End-nodes $\rightarrow$ Gateway $\rightarrow$ Backhaul (4G/WiFi) $\rightarrow$ Cloud Server.
  - **Channels**: **Uplink**: 64 channels (125 kHz) and 8 channels (500 kHz) using frequency hopping. **Downlink**: 8 channels (500 kHz).
  - **ALOHA Access**: Devices transmit whenever they have data; no carrier sensing.
  - **Diversity**: Rely on random channel selection (63 uplink), orthogonality to min. collisions.
- **Constraints & Weaknesses**: high **Latency**: (40ms–1.2s per message); no real-time capability. **Mobility**: Challenging for fast-moving nodes.**Asymmetry**: Primarily one-way (uplink); downlink is severely limited. **Dependency**: Lacks full OSI stack; requires a cloud-based interface/subscription.

13. Cellular Networks

'Cell': Instead of a high-power tower, area divided into smaller Hexagonal Cells.

- **Benefit**: Phones can transmit at much **lower power** because the tower is closer.
- **Frequency Reuse**: Neighboring cells use diff freq (channels) to prevent interference.
- **Generational Evolution of Cellular Networks**:

1G (Analog) & 2G (Digital)

- **1G (AMPS)** - Advanced mobile phone services: Analog voice, frequency modulation.
- **2G (GSM)** - Global System for Mobile Comms: Shift to Digital.
  - **Logic**: Combined **TDMA** (Time Slots) with slow frequency hopping to fight fading.
  - **Features**: Introduced SIM (Subscriber Identity Module) cards, SMS, encryption.
- **2.5G (GPRS)** - General Packet Radio Service: **Packet Switching** for bursty internet data.

3G (CDMA - Code Division Multiple Access), 4G (LTE - Long Term Evolution)

- **CDMA Principle**: Everyone talks at the same time on the same frequency, but each user uses a unique **Orthogonal Code**. (Application of DSSS for multiple access). **Math**: If Code A $\cdot$ Code B = 0, the receiver can 'filter out' the unwanted user.
- **LTE Architecture**: Split into **RAN** (Radio Access Network / eNodeB) and **Core Network**. Uses **OFDMA** and **MIMO**. Completely IP-based, packet-switched network.
- **LTE-Advanced**: Introduced Carrier Aggregation (combining channels for up to 100 MHz).

5G (New Radio)

- **Massive MIMO**: Using arrays with > 64 antennas for fine-grained beamforming.
- **Terahertz/mmWave**: Moving to much higher frequencies for Gbps speeds.

Evolution Summary Table

Gen	Rate	Switching	Access Method
1G	10 Kbps	Circuit	FDMA (Analog)
2G	64 Kbps	Circuit	TDMA / FDMA
3G	2 Mbps	Packet/Circuit	CDMA (DSSS)
4G	100+ Mbps	Packet (IP)	OFDMA / MIMO
5G	10+ Gbps	Packet (IP)	OFDMA / Massive MIMO

Feature	FDMA	TDMA	OFDM
Channel Access	Divides channel into frequency bands, each user assigned a specific frequency	Divides channel into time slots, each user assigned a specific time slot	Uses multiple orthogonal subcarriers simultaneously
Collision Handling	No collisions within allocated frequency bands	No collisions within allocated time slots	Minimizes collisions through orthogonal frequency division
Efficiency	Medium: Bandwidth may be wasted if user has no data to send	Medium-High: Time slots may be wasted if user has no data to send	High: Efficient spectrum usage through parallel transmissions
Interference	Vulnerable to narrowband interference	Less vulnerable to frequency-specific interference	Resistant to narrowband interference and multipath fading
Synchronization	No tight synchronization needed	Requires precise time synchronization	Requires frequency synchronization
Best Used In	Simple constant-bit-rate applications, analog systems	Digital cellular systems, satellite communications	Modern wireless networks (Wi-Fi, 4G/5G), high-speed data transmission

Feature	Aloha	Slotted Aloha	CSMA
Channel Access	Transmits immediately when data is ready without checking channel status	Transmits only at beginning of predetermined time slots	Listens to the channel before transmitting (sense-before-send)
Collision Handling	Collisions occur frequently due to uncoordinated transmissions	Reduced collisions compared to pure Aloha, but still occur	Significantly reduces collisions by avoiding transmission when channel is busy
Retransmission	Random backoff time before retrying after collision	Random waiting time before retrying, synchronized to slots	Uses various backoff algorithms (e.g., exponential) to reduce subsequent collisions
Efficiency	Very low: 18% of channel capacity	Low-Medium: 36% of channel capacity	Medium-High: Up to 80% in optimal conditions
Collision Avoidance	None; simply retransmits after collision	None; retransmits at next available slot	Various mechanisms like RTS/CTS and collision detection depending on variant
Best Used In	Simple, very low-traffic networks	Networks with predictable traffic patterns	Wireless and wired LANs with variable traffic loads

Feature	ALOHA with preamble	B-MAC	X-MAC
Channel Access	Sends a preamble before data transmission to announce intent	Uses Low Power Listening (LPL) with long preambles	Uses shortened preambles with target address
Collision Handling	Reduced collisions compared to pure Aloha	Collisions reduced through channel sensing	Improved collision avoidance through early acknowledgment
Power Efficiency	Low: Continuous listening	Medium: Periodic channel sampling	High: Early acknowledgment reduces overhearing
Latency	Medium	High due to long preamble	Reduced compared to B-MAC
Adaptability	Limited adaptability to traffic patterns	Configurable sleep intervals	Adaptable to varying traffic patterns
Best Used In	Simple networks with low power constraints	Wireless sensor networks with low duty cycles	Energy-constrained sensor networks with variable traffic

References and Credits

- Course Content topics, ordering from CS4222, National University of Singapore.
- Select contents, diagrams referenced from @JasonYapzx CS4222 Cheatsheet